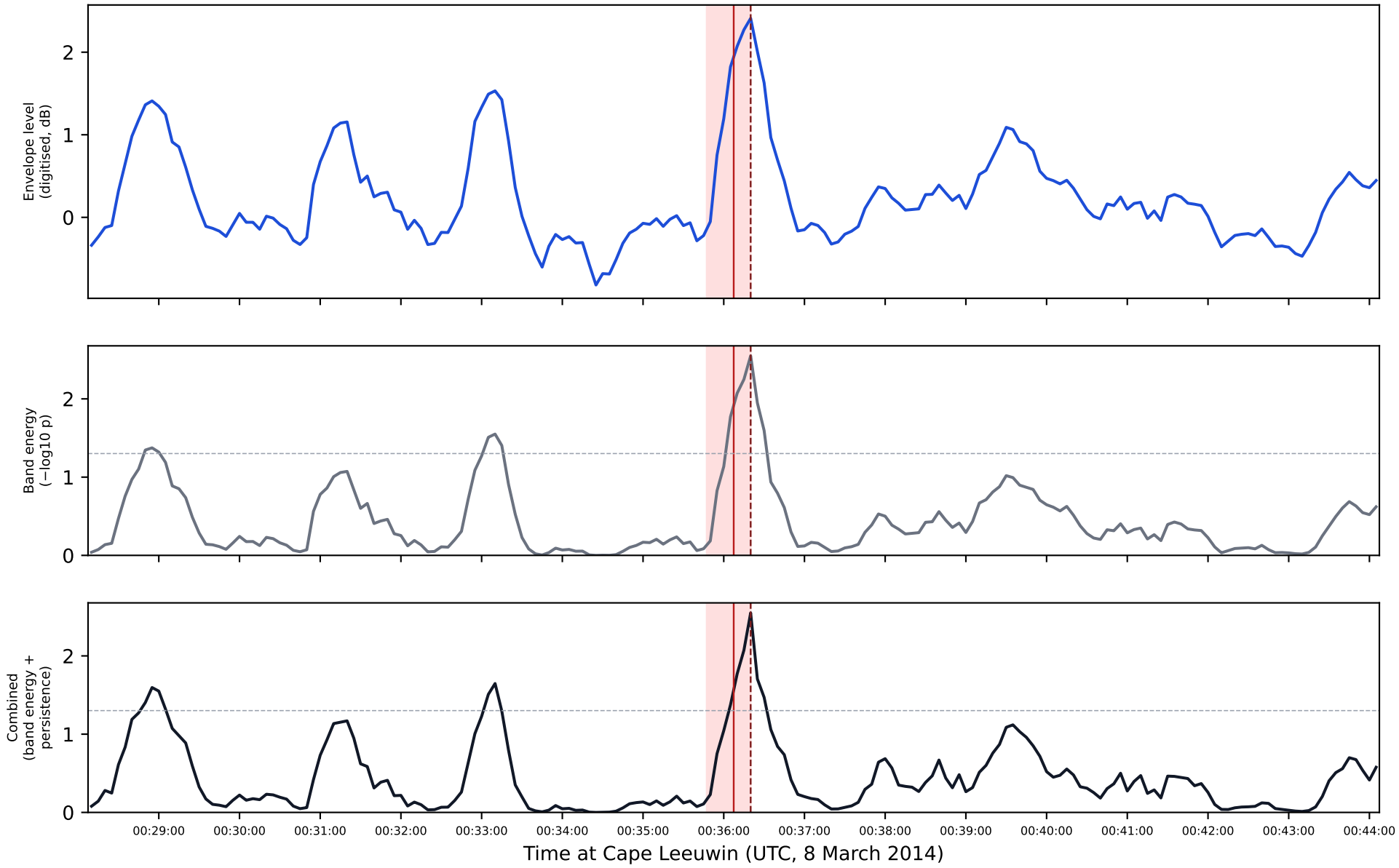
Ten strongest Cape Leeuwin and Diego Garcia South candidates with an implied source on the 7th arc

Rank	Station	Arrival	Bearing, published / corrected	Found by	Envelope p	Arc latitude (° S)	Implied impact	Steep dive	Time coincidence (label only)
1	Cape Leeuwin	00:36:07	none	ours	0.003	any (no bearing)	00:19:38–00:19:44	yes	Royer B, figure pick +0 s
2	Cape Leeuwin	00:47:04	343.2° / 338.7° (±2.0°)	Kadri and ours	0.006	15; 16 (outside 20-41)	00:19:38–00:24:50	yes	
3	Diego Garcia S.	01:10:15	none	ours	0.009	any (no bearing)	00:24:02–00:33:59	yes	
4	Cape Leeuwin	00:54:30	306.2° / 299.4° (±3.9°)	Kadri and ours	0.062	27	00:32:59–00:40:58	no	Royer 5 (uncertain), table pick -18 s
5	Cape Leeuwin	00:51:50	320.4° / 315.5° (±3.9°)	Kadri and ours	0.096	23; 24	00:29:14–00:37:50	no	Royer A, figure pick +7 s
6	Cape Leeuwin	00:49:58	260.4° / 246.8° (±2.1°)	Kadri	0.121	40	00:20:59–00:21:26	yes	
7	Cape Leeuwin	00:55:07	314.1° / 308.5° (±3.9°)	Kadri	0.237	25; 26	00:32:59–00:41:11	no	
8	Cape Leeuwin	00:45:36	312.0° / 306.1° (±3.9°)	Kadri	0.273	25; 26	00:25:09–00:30:42	yes	
9	Cape Leeuwin	00:40:34	303.3° / 296.0° (±3.9°)	Kadri	0.332	28	00:21:13–00:25:18	yes	
10	Cape Leeuwin	00:41:43	310.0° / 303.8° (±3.9°)	Kadri	0.344	26; 27	00:22:07–00:26:03	yes	

Ranking: envelope evidence at the station itself (band energy and persistence on the digitised Kadri (2024) Figure 9 envelope, smallest p within ±15 s; within-trace null). This is weak evidence: the p floor is about 1/355 and the 19 Kadri times are not corrected for multiple looks. Sources: Kadri Table 1; ours = combined-detector best looks on the envelopes and 4 Nov screening picks (merged within 30 s of a Kadri time, or 45 s of each other). Bearings: Kadri's published bearings corrected for the H01W array-position bias (calibration from 144 earthquakes, ruling B1; corrections -4.4 to -13.8°); tolerance ±1.645σ with σ = calibrated sector error and a 1.2° weak-signal term. Stronger candidates not shown: 00:39:53 (published 256.8°, corrected 243.0°; envelope p 0.093): bearing fits 40-43 S, but implied impact 23:57:02-00:12:19 is before 00:19:38; 00:49:41 (published 234.7°, corrected 221.7°; envelope p 0.121): corrected bearing meets no 7th-arc point; 00:38:29 (published 261.2°, corrected 247.7°; envelope p 0.214): bearing fits 38-43 S, but implied impact 23:55:06-00:15:19 is before 00:19:38; 00:39:02 (published 268.2°, corrected 255.3°; envelope p 0.251): bearing fits 36-40 S, but implied impact 00:05:12-00:19:30 is before 00:19:38. Implied impact: arc points 15-43° S within the bearing tolerance (no bearing: 20-41° S), sound speed: physical bounds per path (WOA23, 5 % detour) together with Kadri's 1.45 to 1.57 km/s, arrival span from the onset minus 20 s to the envelope peak, inside 00:19:38 to 00:55:00. Steep dive: impact 00:19:38 to 00:29:44 within 25 NM. Catalogue labels are associations below 0.95 probability (label only, not removed).

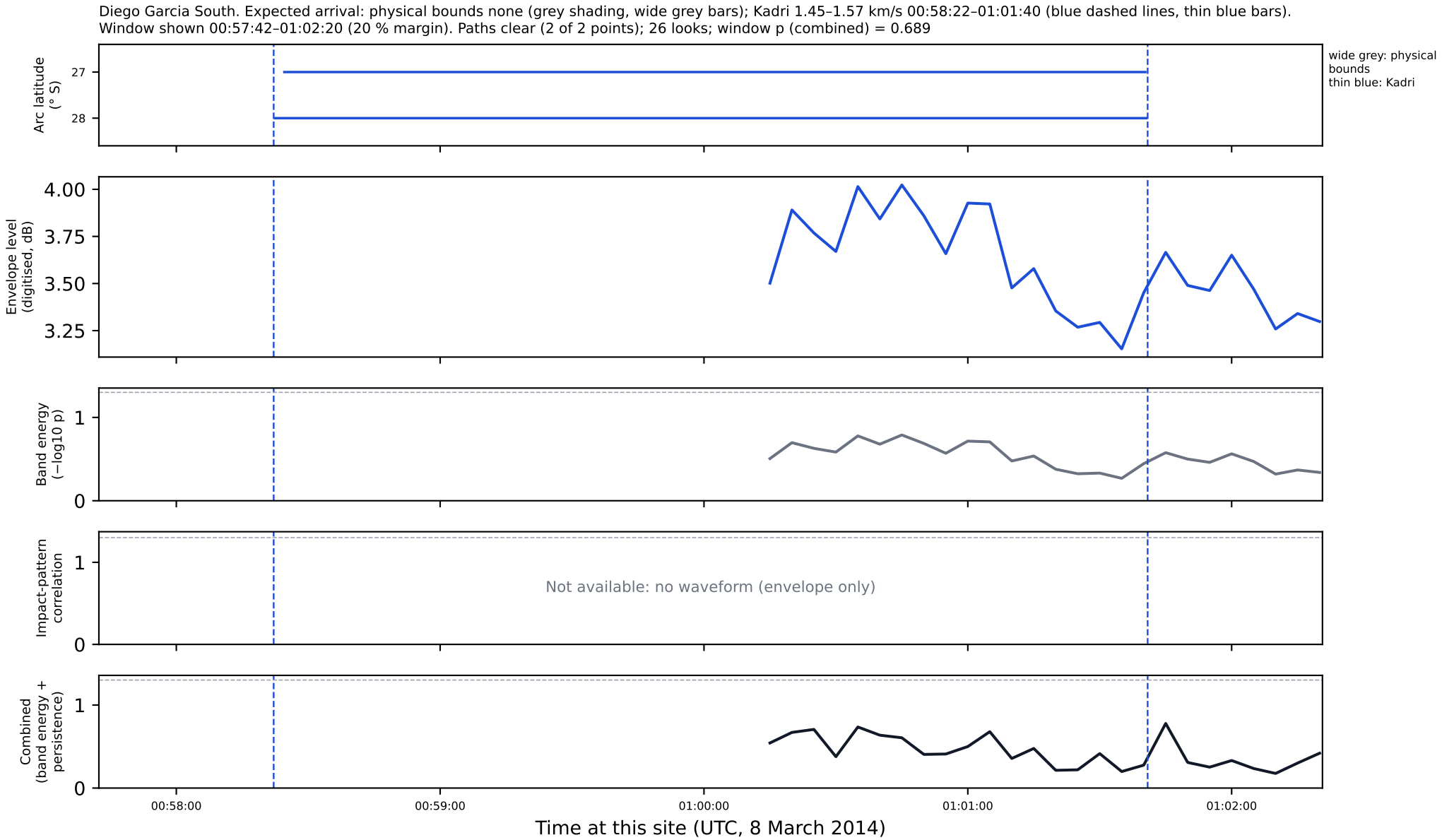
Candidate 1: Cape Leeuwin 00:36:07 (ours; no bearing). Implied impact 00:19:38–00:19:44 UTC, arc latitude from timing only

Primary site, Cape Leeuwin. Solid red line: onset pick 00:36:07. Dashed line: envelope peak 00:36:20. Red band: arrival span used for timing, 00:35:47–00:36:20.
Envelope evidence p = 0.00282 (smallest within ±15 s of the onset; within-trace null).
Coincides in time with Royer B, figure pick +0 s; Royer B (uncertain), table pick -13 s (label only, not removed; Royer's event locations rest on his own unverified pick association).



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

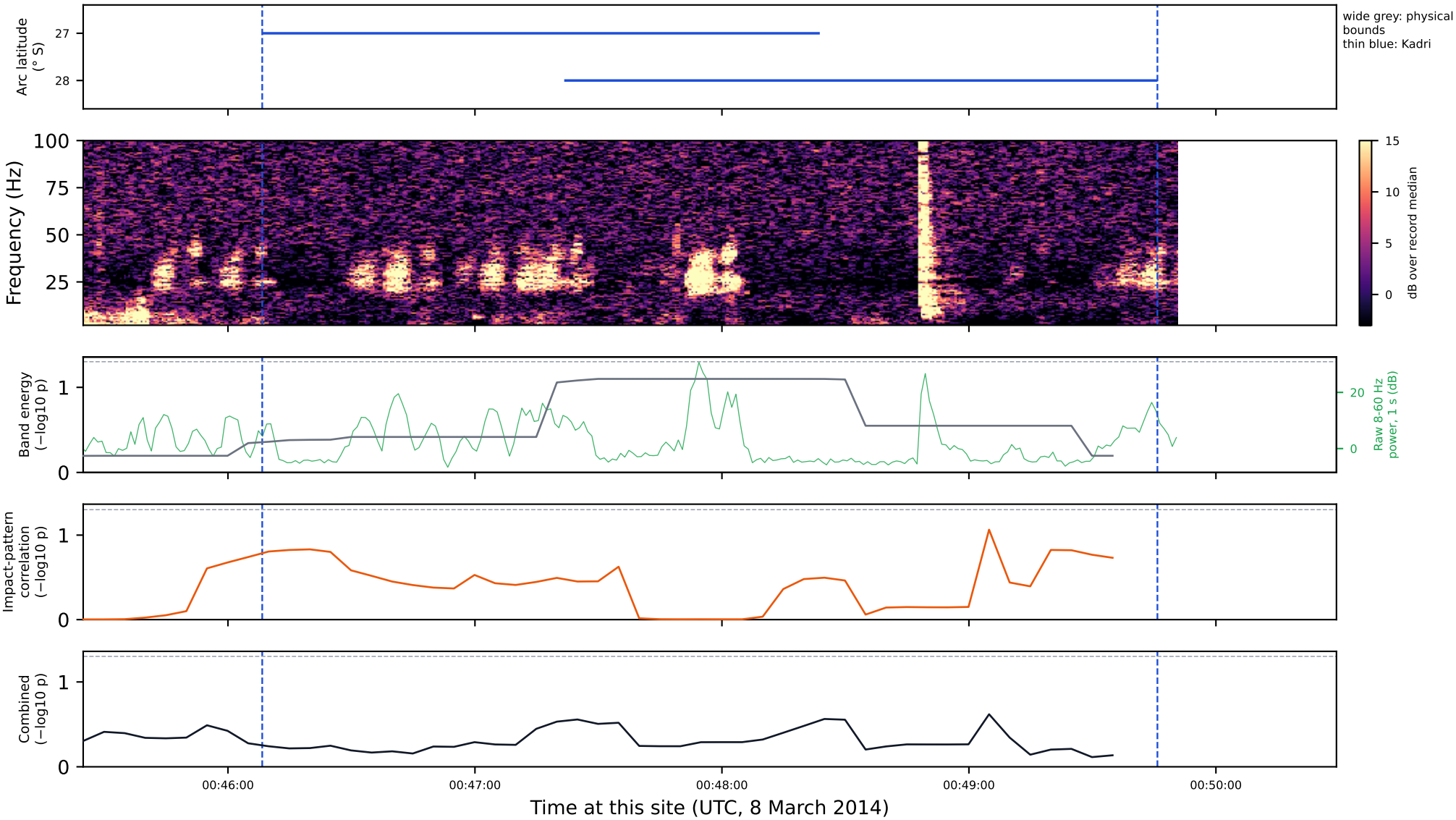
Candidate 1: Cape Leeuwin 00:36:07 (ours; no bearing). Implied impact 00:19:38–00:19:44 UTC, arc latitude from timing only



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

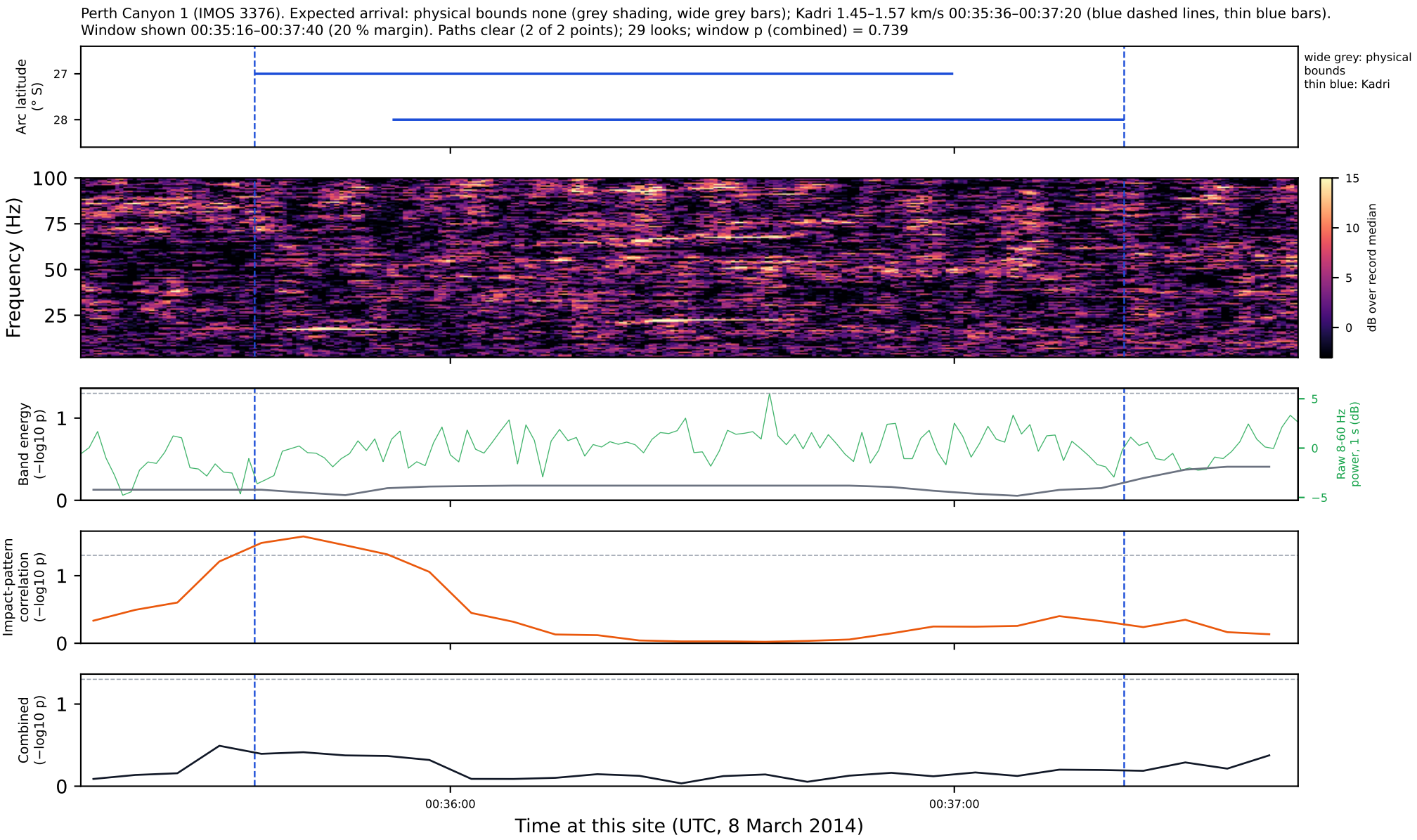
Candidate 1: Cape Leeuwin 00:36:07 (ours; no bearing). Implied impact 00:19:38–00:19:44 UTC, arc latitude from timing only

Scott Reef (IMOS 3250). Expected arrival: physical bounds none (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:46:08–00:49:45 (blue dashed lines, thin blue bars).
Window shown 00:45:24–00:50:29 (20 % margin). Paths large attenuation (2 of 2 points); 51 looks; window p (combined) = 0.779



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

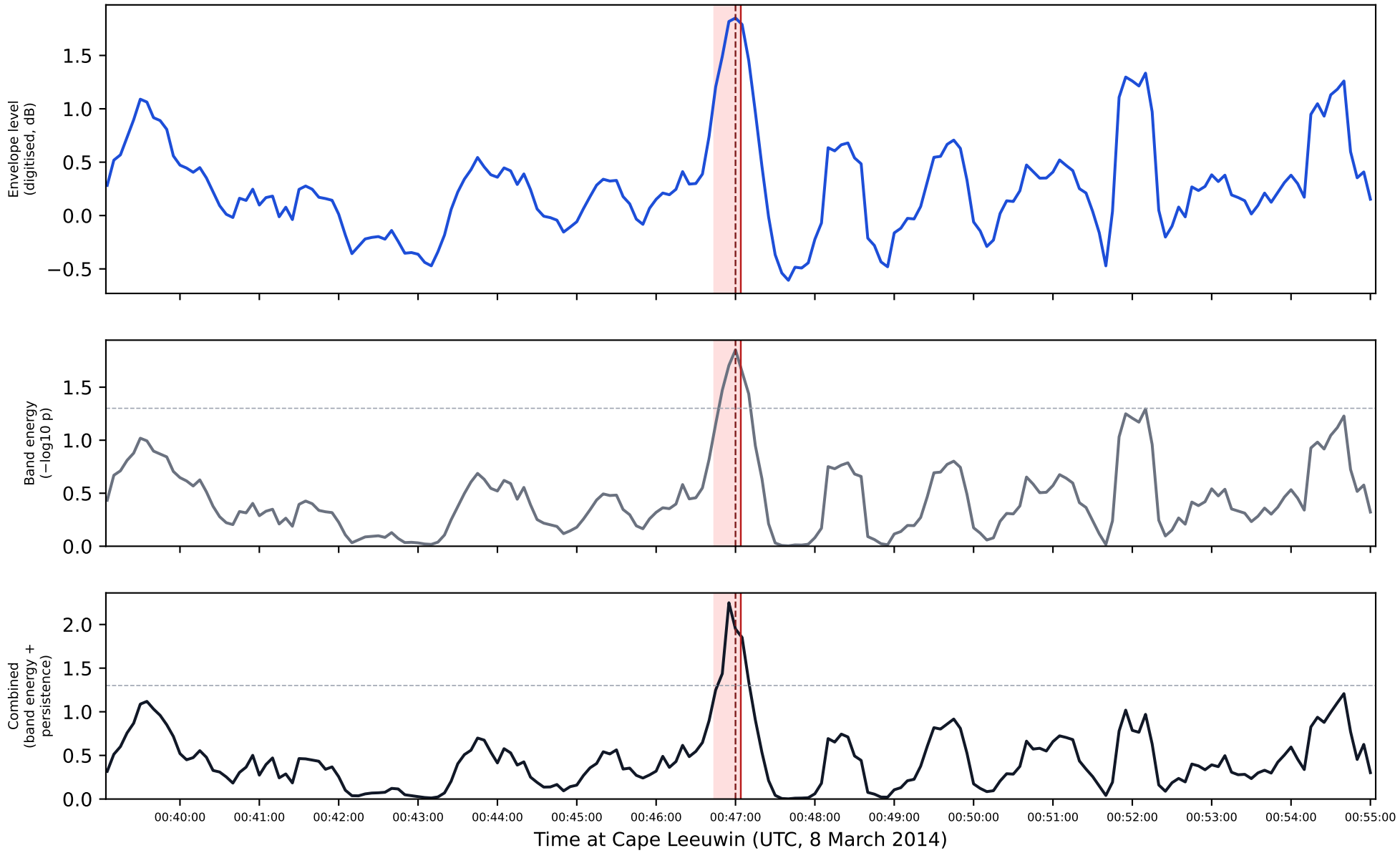
Candidate 1: Cape Leeuwin 00:36:07 (ours; no bearing). Implied impact 00:19:38–00:19:44 UTC, arc latitude from timing only



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

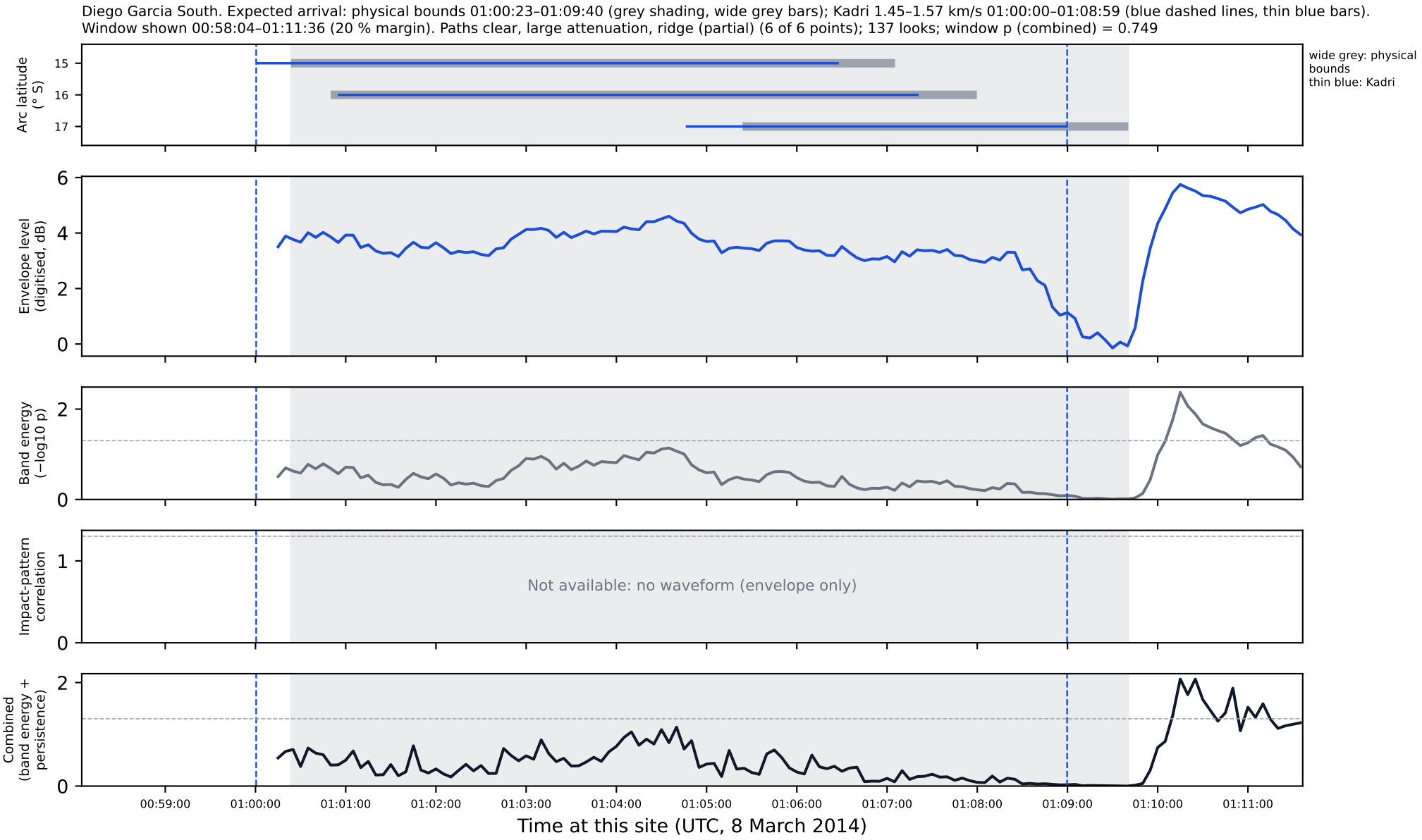
Candidate 2: Cape Leeuwin 00:47:04 (Kadri and ours; bearing 338.7° corrected, 343.2° published). Implied impact 00:19:38–00:24:50 UTC, arc 15; 16° S

Primary site, Cape Leeuwin. Solid red line: onset pick 00:47:04. Dashed line: envelope peak 00:47:00. Red band: arrival span used for timing, 00:46:44–00:47:04.
Envelope evidence p = 0.00563 (smallest within ±15 s of the onset; within-trace null).



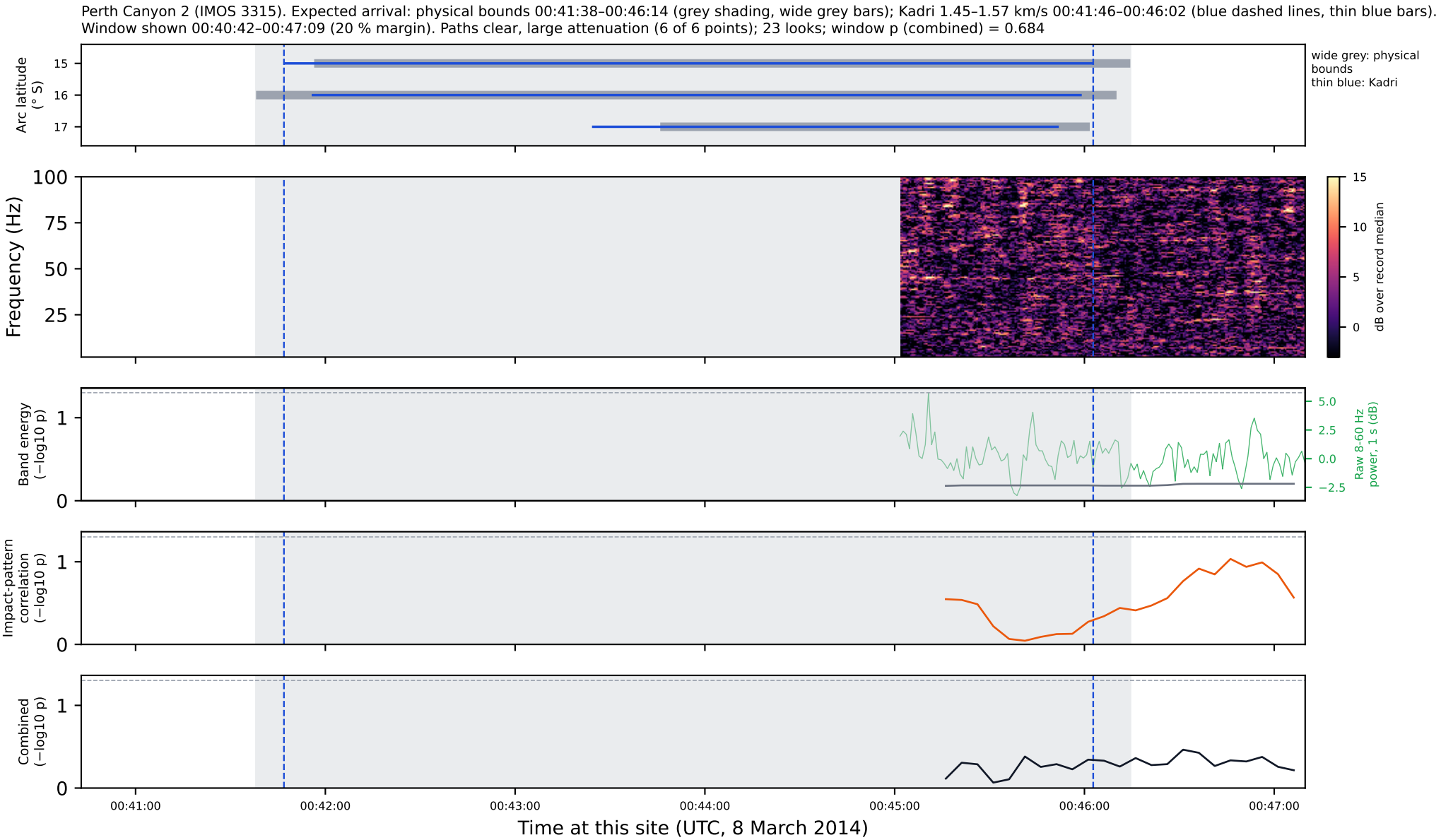
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

Candidate 2: Cape Leeuwin 00:47:04 (Kadri and ours; bearing 338.7° corrected, 343.2° published). Implied impact 00:19:38–00:24:50 UTC, arc 15; 16° S



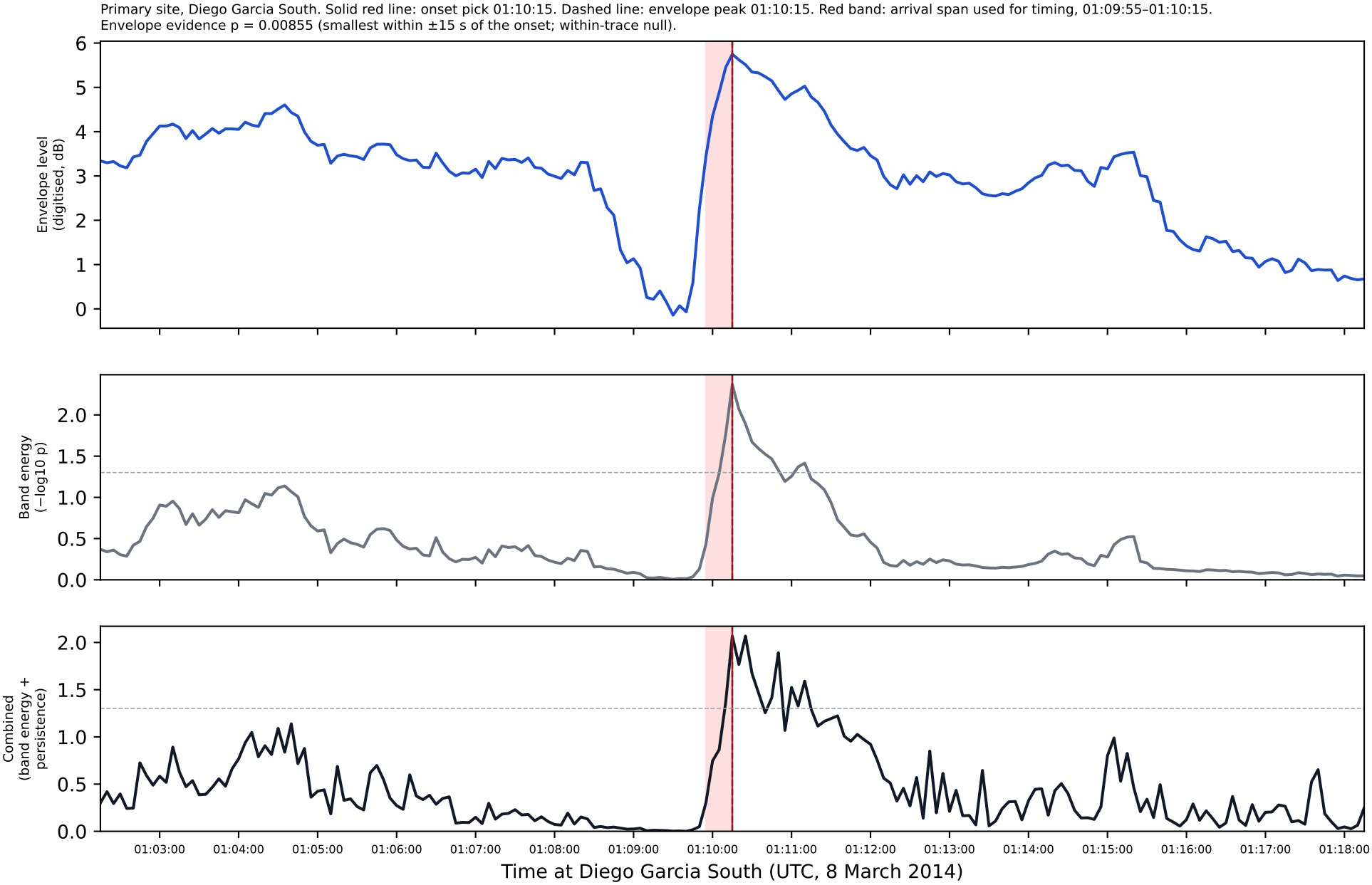
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 2: Cape Leeuwin 00:47:04 (Kadri and ours; bearing 338.7° corrected, 343.2° published). Implied impact 00:19:38–00:24:50 UTC, arc 15; 16° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

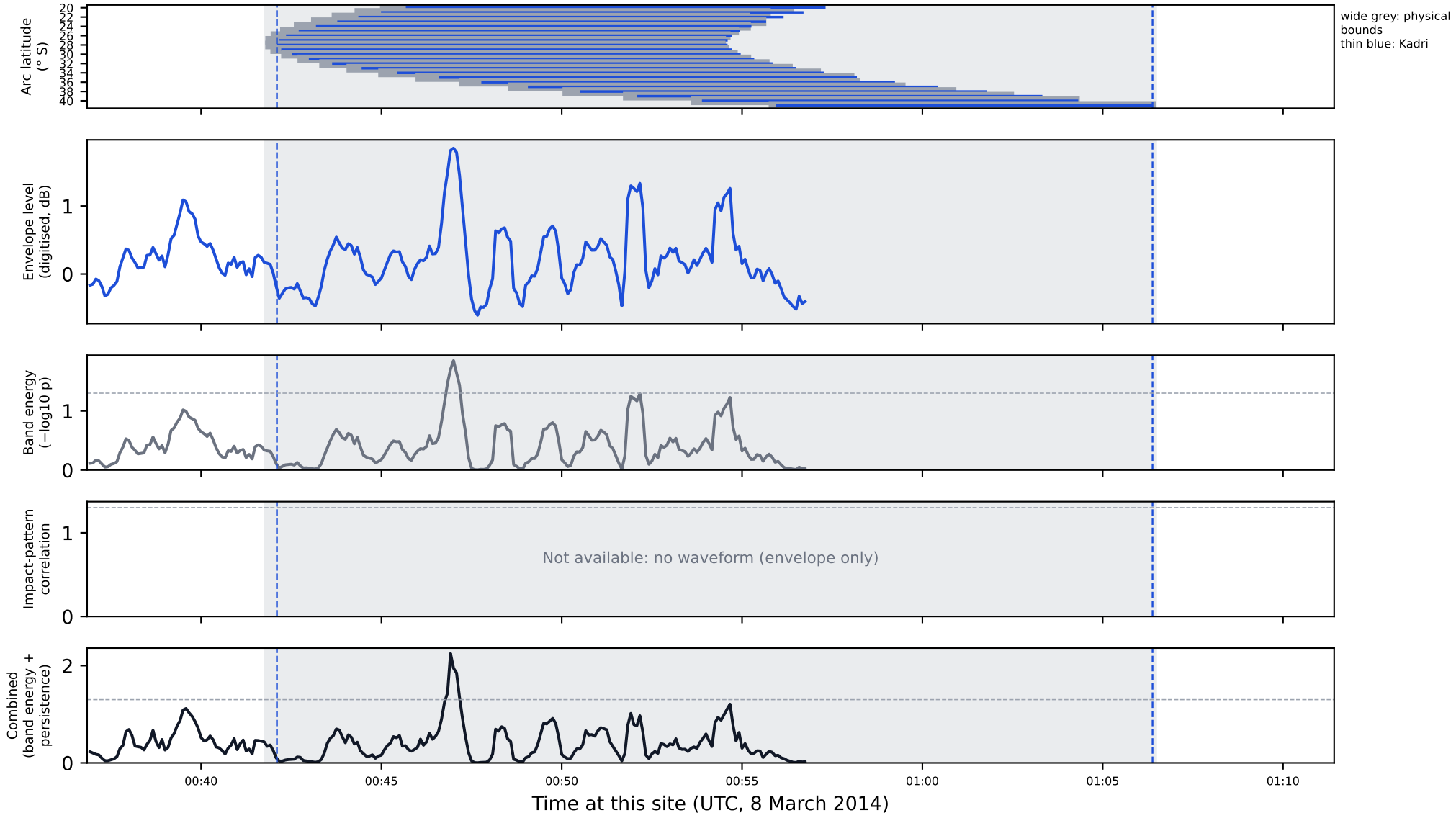
Candidate 3: Diego Garcia South 01:10:15 (ours; no bearing). Implied impact 00:24:02–00:33:59 UTC, arc latitude from timing only



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

Candidate 3: Diego Garcia South 01:10:15 (ours; no bearing). Implied impact 00:24:02–00:33:59 UTC, arc latitude from timing only

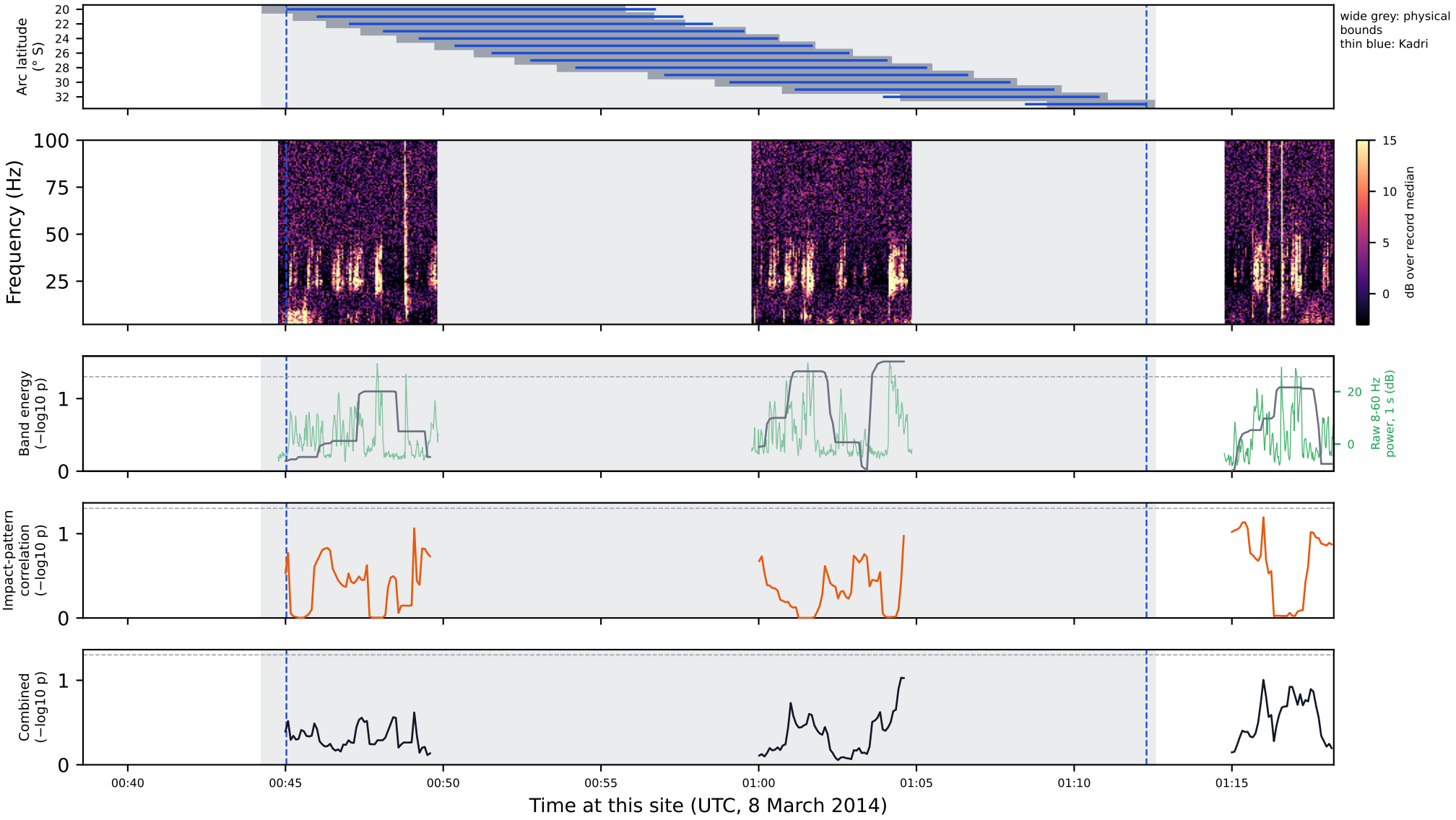
Cape Leeuwin. Expected arrival: physical bounds 00:41:46–01:06:28 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:42:06–01:06:22 (blue dashed lines, thin blue bars).
Window shown 00:36:50–01:11:25 (20 % margin). Paths clear, large attenuation, ridge (partial) (196 of 196 points); 239 looks; window p (combined) = 0.786



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 3: Diego Garcia South 01:10:15 (ours; no bearing). Implied impact 00:24:02–00:33:59 UTC, arc latitude from timing only

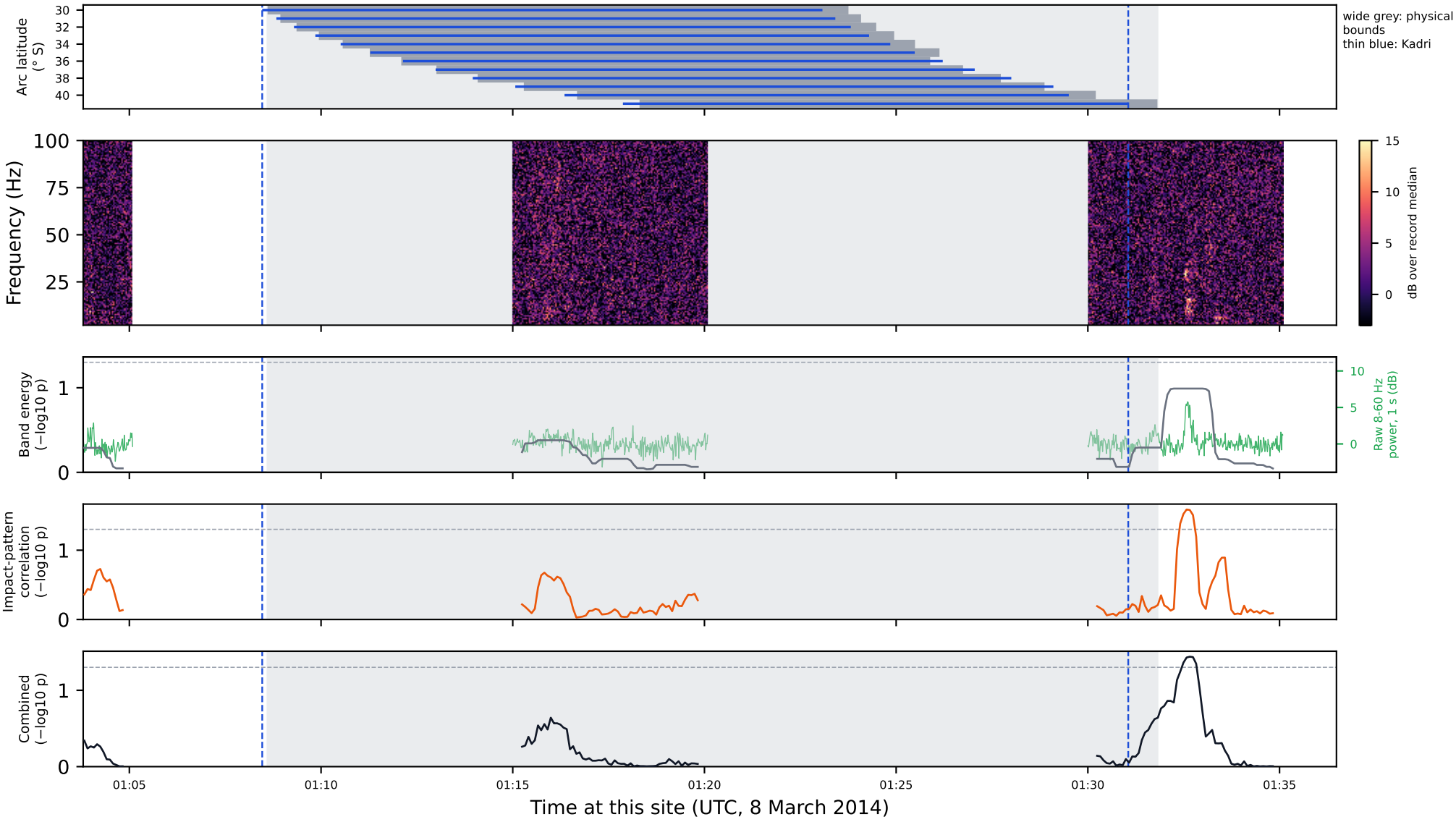
Scott Reef (IMOS 3250). Expected arrival: physical bounds 00:44:14–01:12:33 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:45:01–01:12:17 (blue dashed lines, thin blue bars).
Window shown 00:38:34–01:18:13 (20 % margin). Paths large attenuation (98 of 196 points); 151 looks; window p (combined) = 0.444



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 3: Diego Garcia South 01:10:15 (ours; no bearing). Implied impact 00:24:02–00:33:59 UTC, arc latitude from timing only

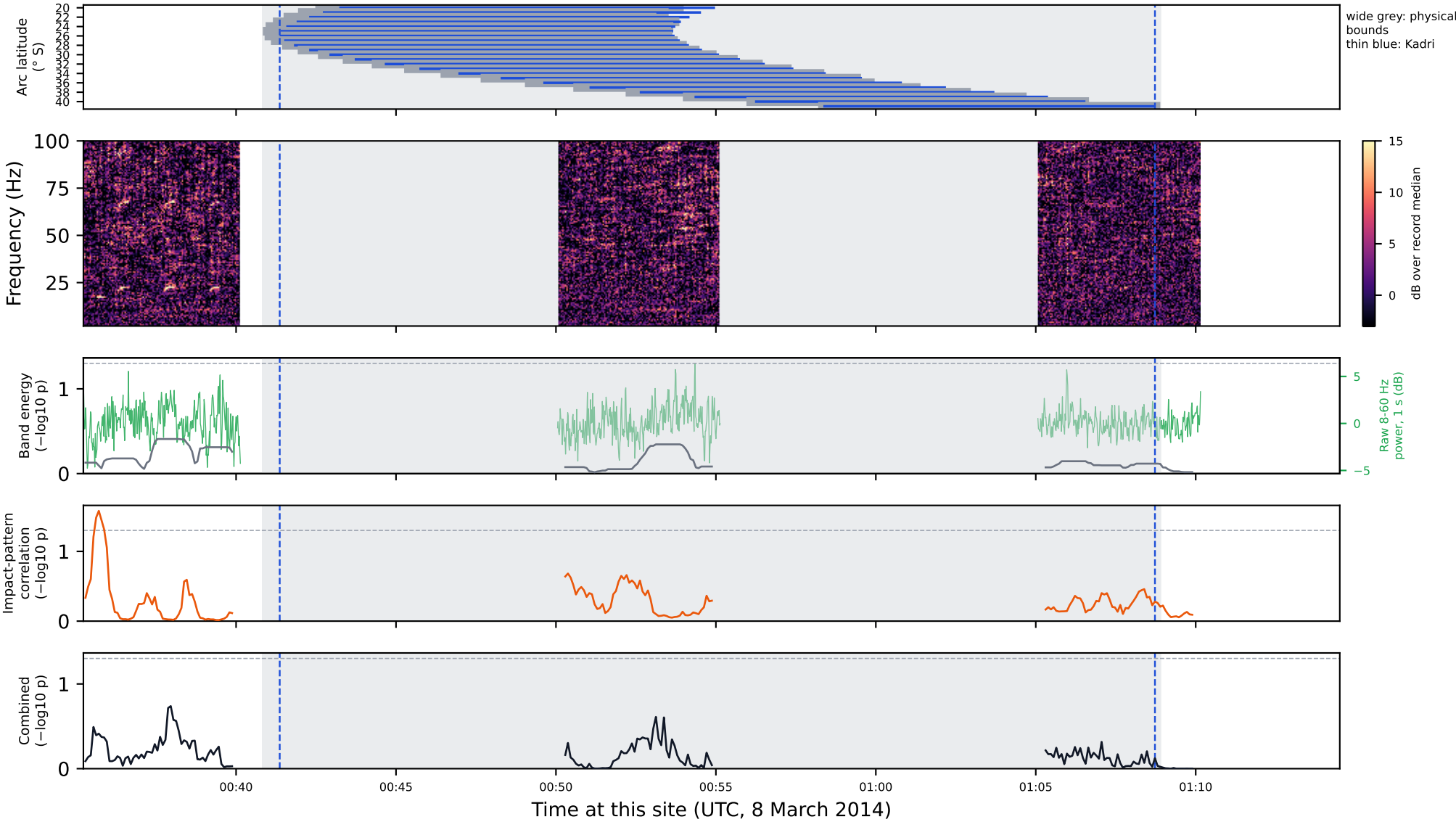
Portland 1 (IMOS 3274). Expected arrival: physical bounds 01:08:36–01:31:48 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 01:08:27–01:31:03 (blue dashed lines, thin blue bars).
Window shown 01:03:47–01:36:29 (20 % margin). Paths clear, large attenuation (106 of 196 points); 125 looks; window p (combined) = 0.249



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 3: Diego Garcia South 01:10:15 (ours; no bearing). Implied impact 00:24:02–00:33:59 UTC, arc latitude from timing only

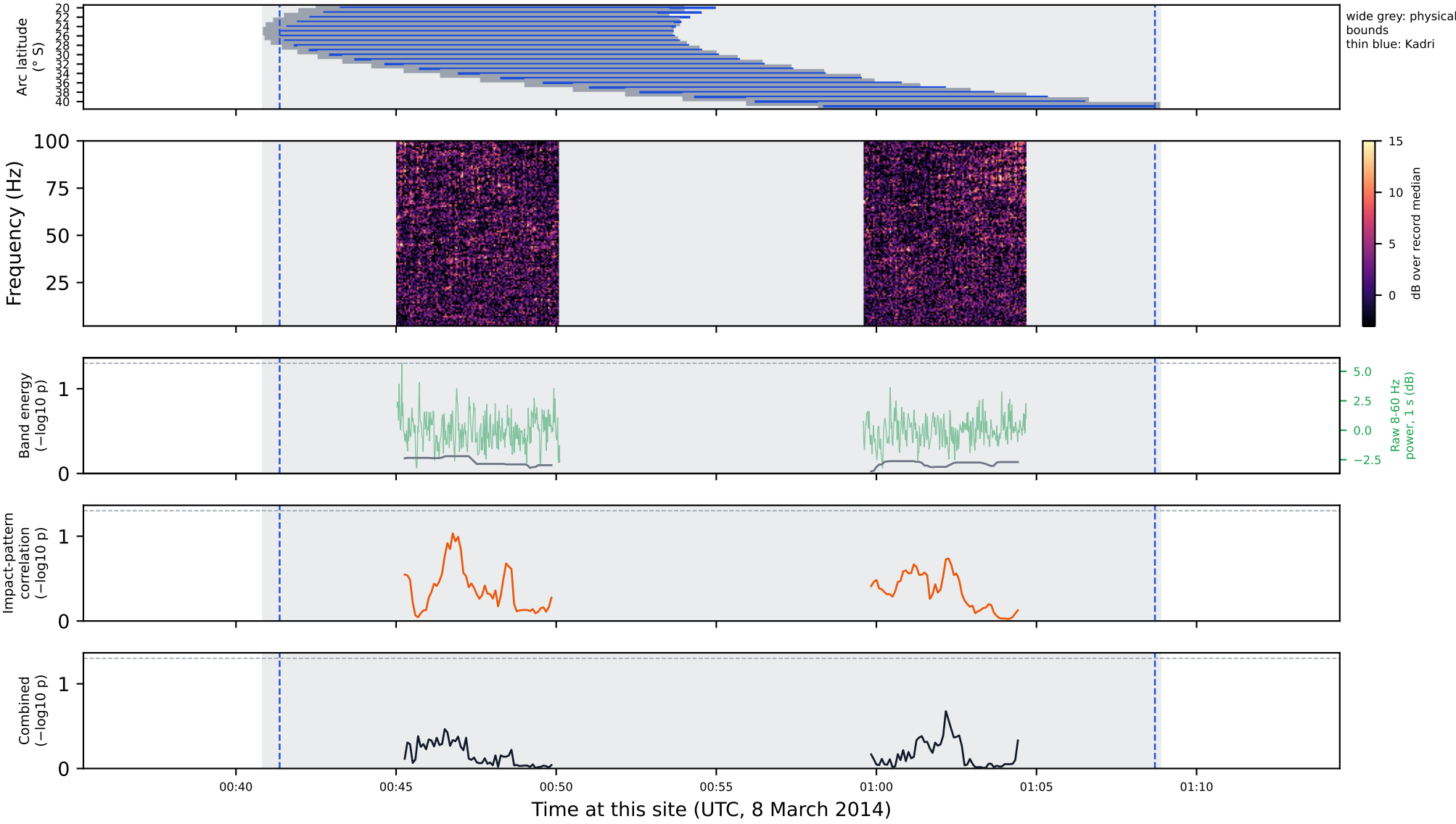
Perth Canyon 1 (IMOS 3376). Expected arrival: physical bounds 00:40:50–01:08:53 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:41:21–01:08:43 (blue dashed lines, thin blue bars).
Window shown 00:35:13–01:14:30 (20 % margin). Paths clear, large attenuation (196 of 196 points); 168 looks; window p (combined) = 0.708



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

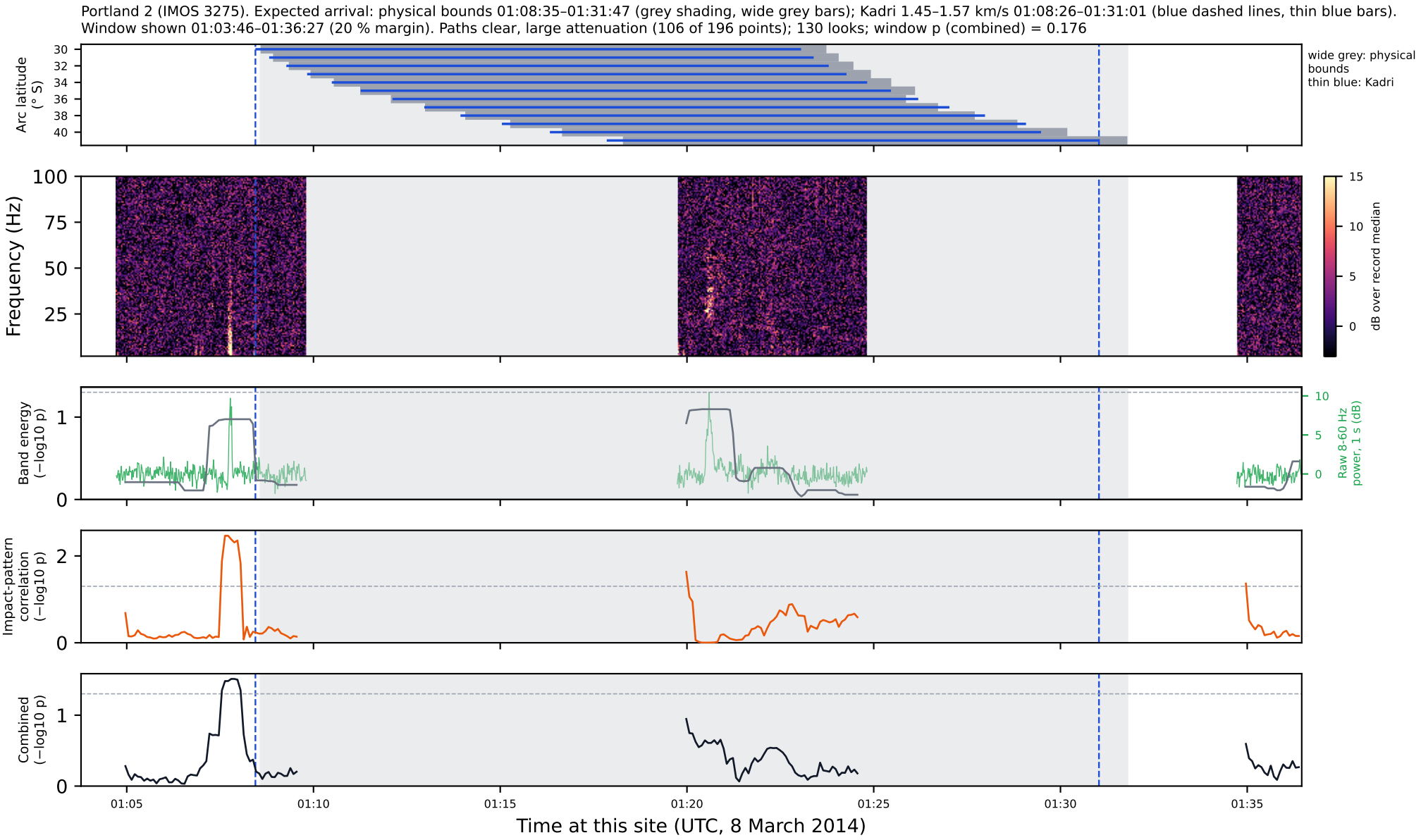
Candidate 3: Diego Garcia South 01:10:15 (ours; no bearing). Implied impact 00:24:02–00:33:59 UTC, arc latitude from timing only

Perth Canyon 2 (IMOS 3315). Expected arrival: physical bounds 00:40:50–01:08:52 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:41:21–01:08:42 (blue dashed lines, thin blue bars).
Window shown 00:35:14–01:14:28 (20 % margin). Paths clear, large attenuation (196 of 196 points); 112 looks; window p (combined) = 0.776



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

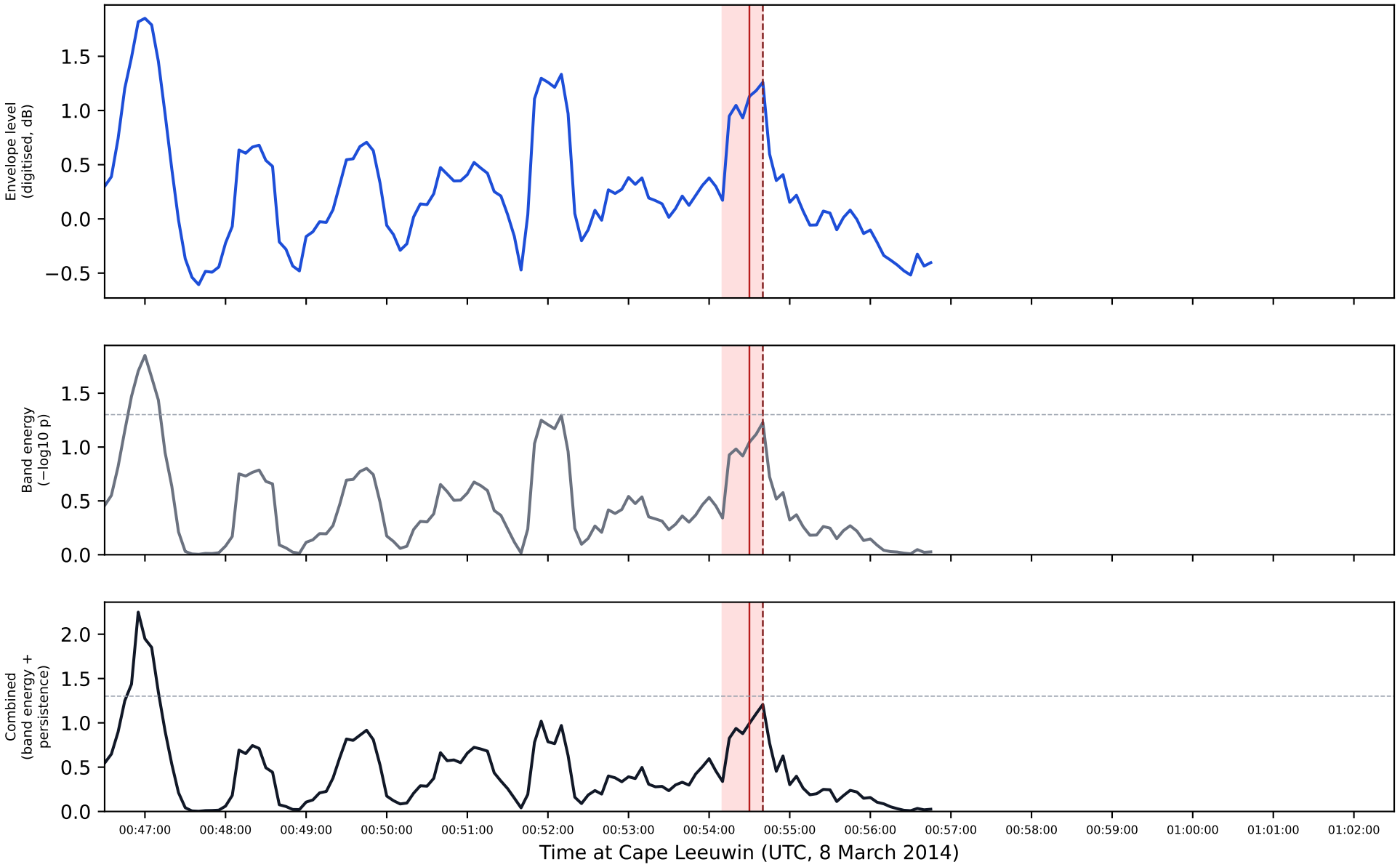
Candidate 3: Diego Garcia South 01:10:15 (ours; no bearing). Implied impact 00:24:02–00:33:59 UTC, arc latitude from timing only



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

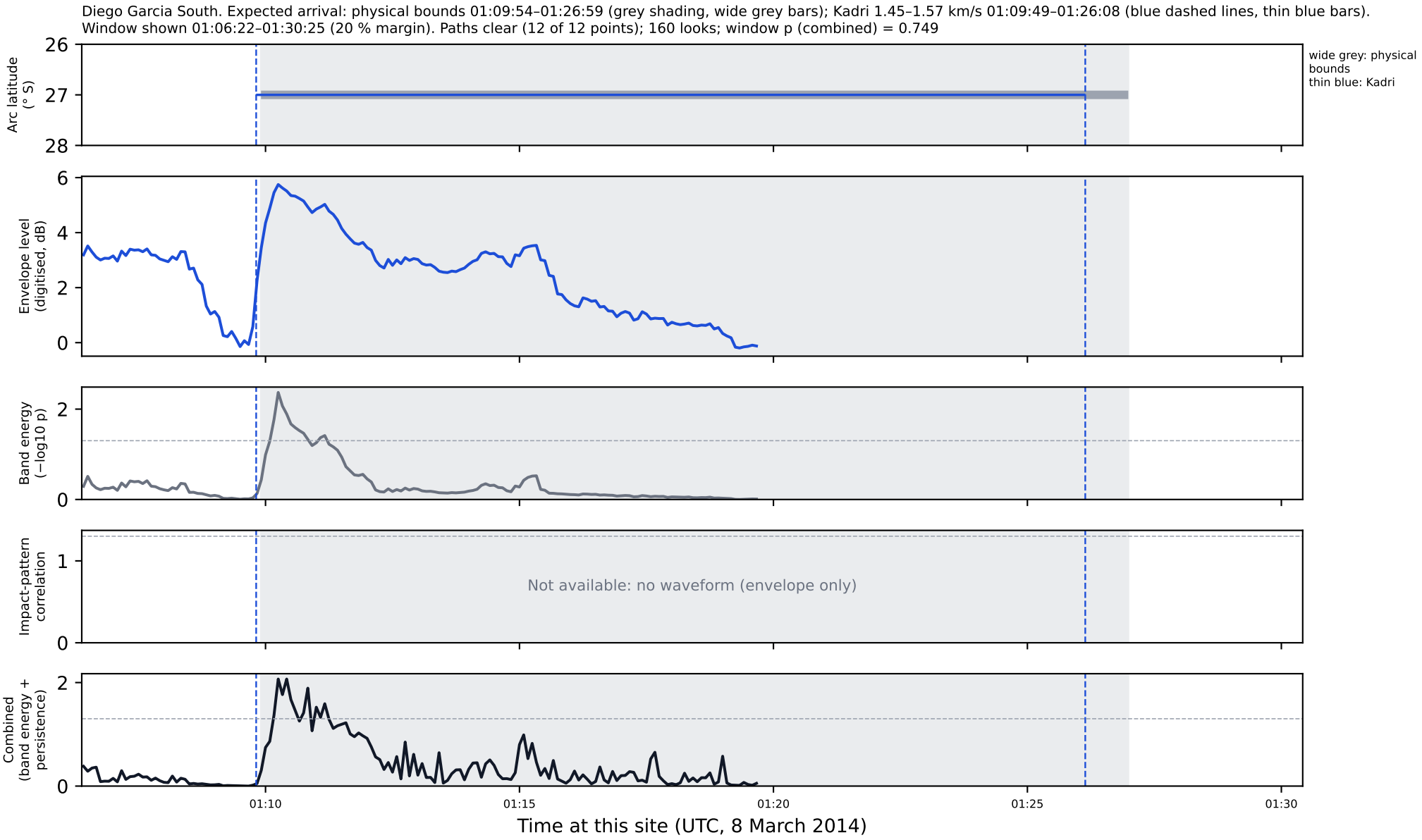
Candidate 4: Cape Leeuwin 00:54:30 (Kadri and ours; bearing 299.4° corrected, 306.2° published). Implied impact 00:32:59–00:40:58 UTC, arc 27° S

Primary site, Cape Leeuwin. Solid red line: onset pick 00:54:30. Dashed line: envelope peak 00:54:40. Red band: arrival span used for timing, 00:54:10–00:54:40.
Envelope evidence p = 0.062 (smallest within ±15 s of the onset; within-trace null).
Coincides in time with Royer 5 (uncertain), table pick -18 s; Royer 5 (uncertain), table pick -20 s (label only, not removed; Royer's event locations rest on his own unverified pick association).



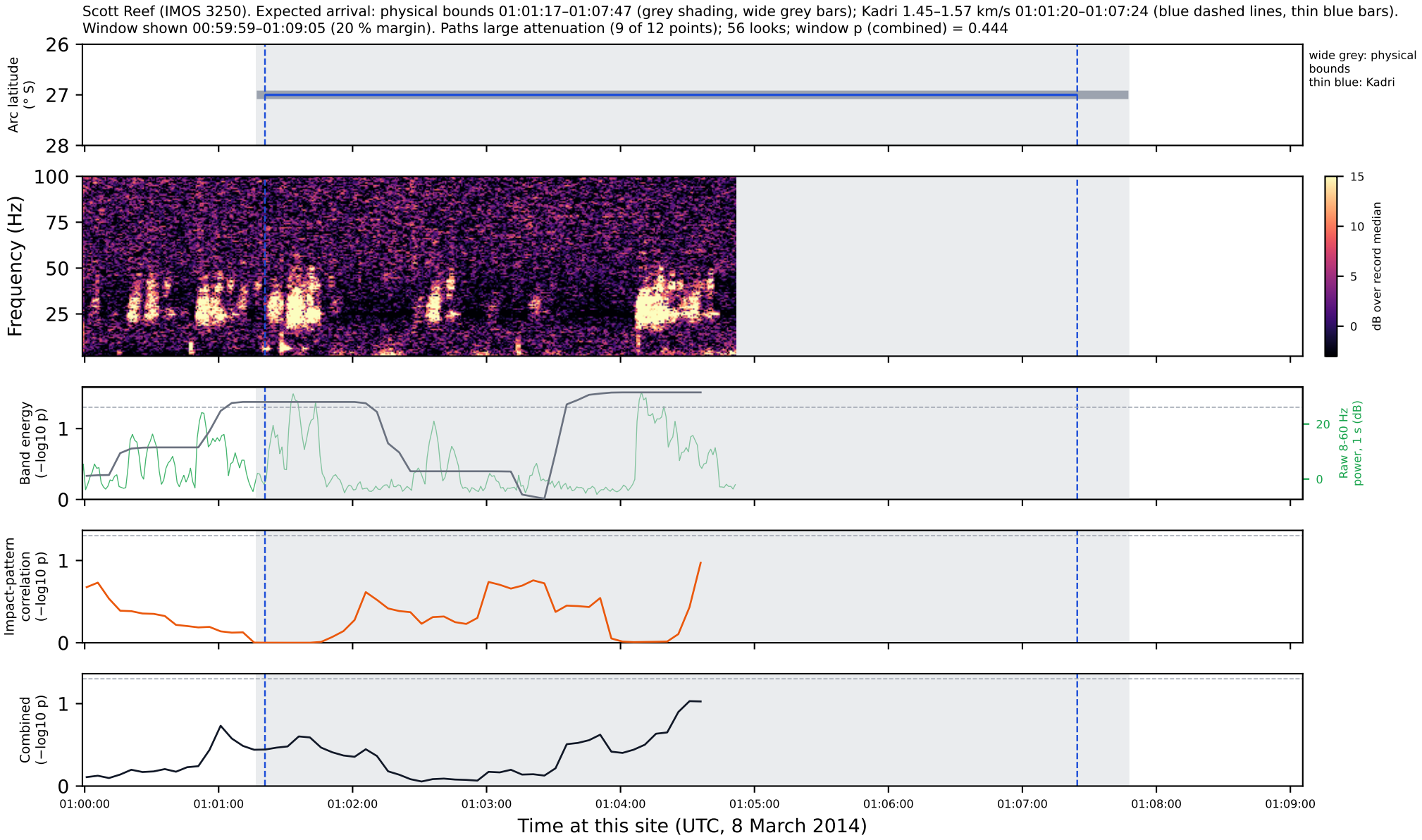
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

Candidate 4: Cape Leeuwin 00:54:30 (Kadri and ours; bearing 299.4° corrected, 306.2° published). Implied impact 00:32:59–00:40:58 UTC, arc 27° S



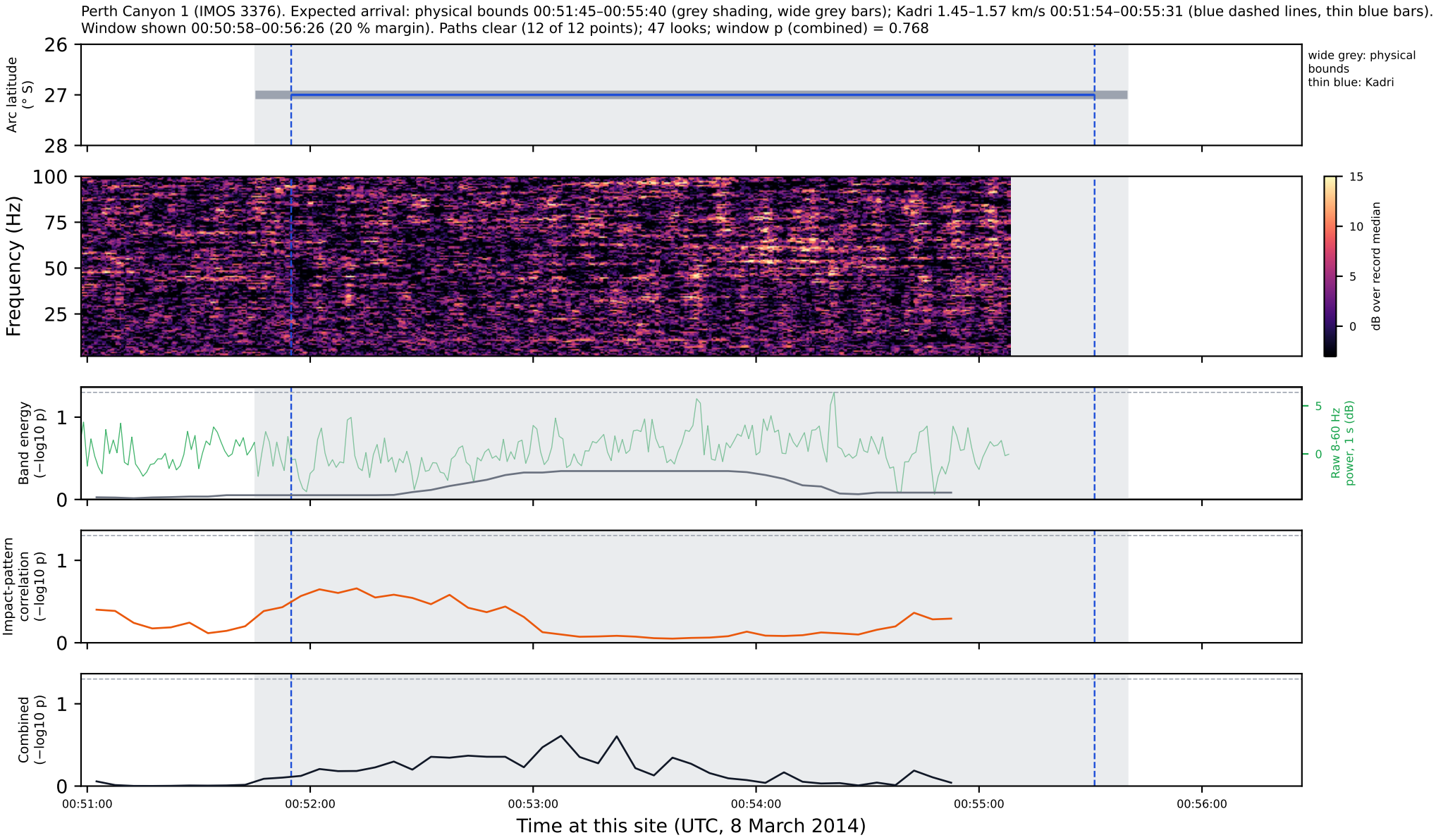
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 4: Cape Leeuwin 00:54:30 (Kadri and ours; bearing 299.4° corrected, 306.2° published). Implied impact 00:32:59–00:40:58 UTC, arc 27° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

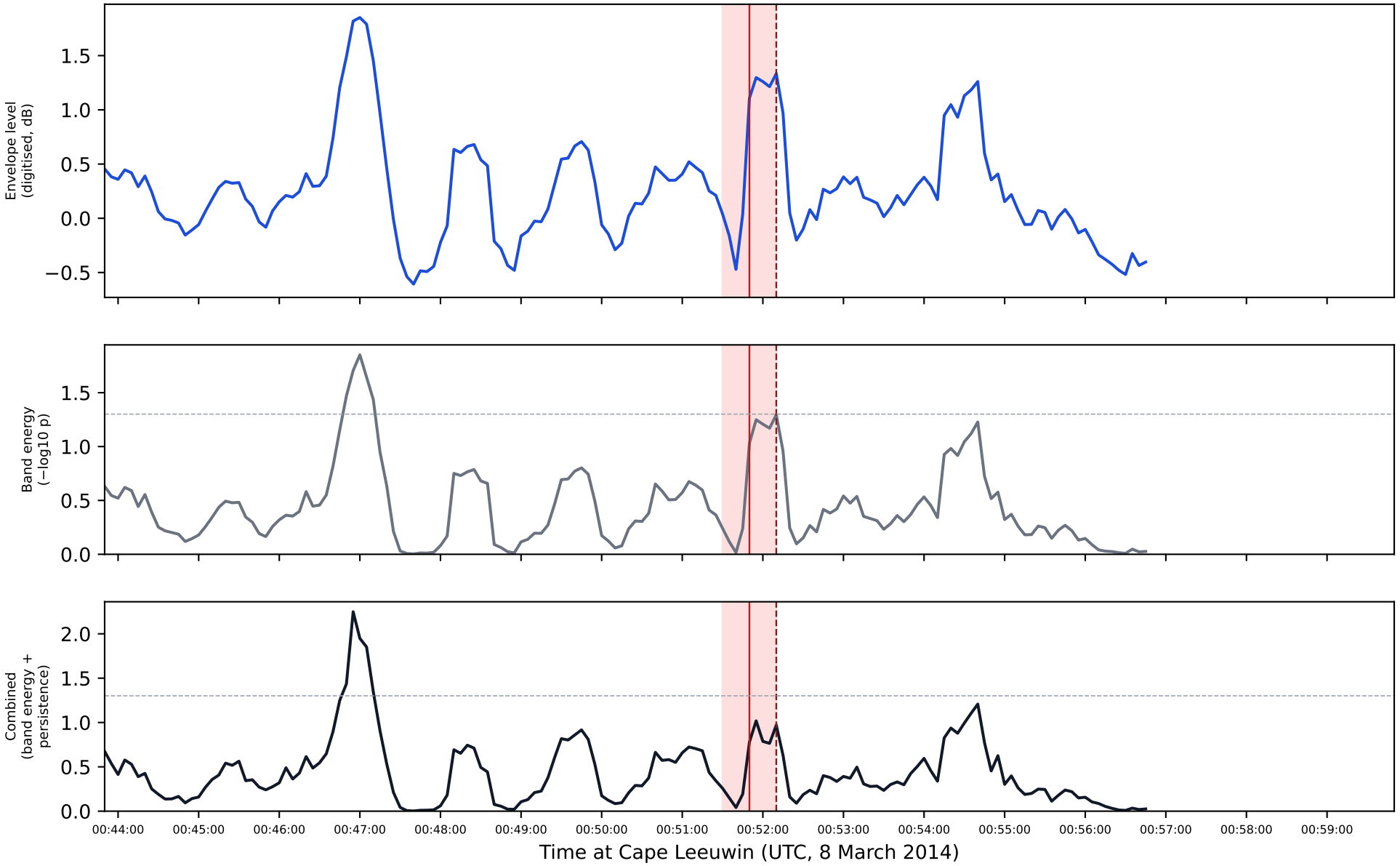
Candidate 4: Cape Leeuwin 00:54:30 (Kadri and ours; bearing 299.4° corrected, 306.2° published). Implied impact 00:32:59–00:40:58 UTC, arc 27° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

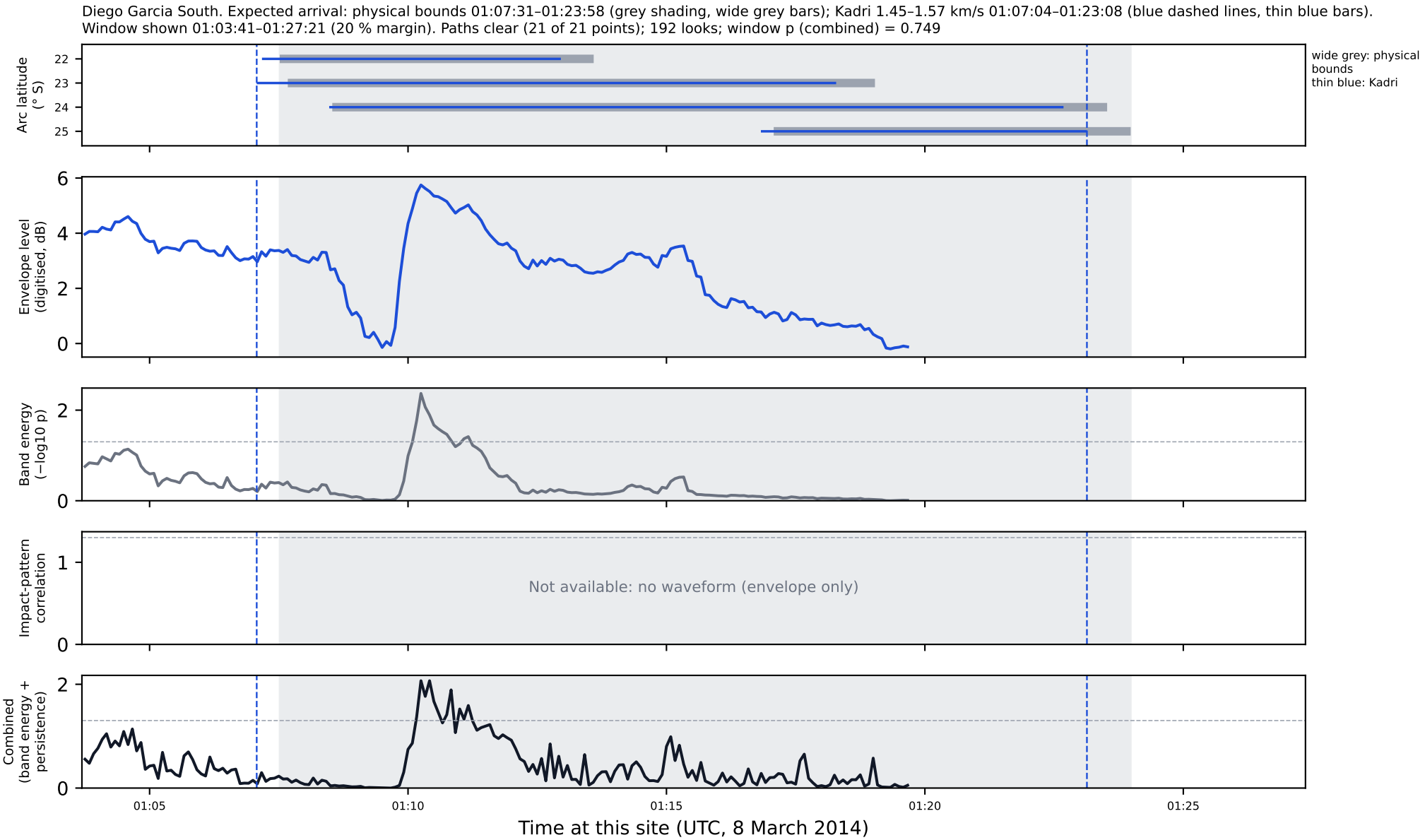
Candidate 5: Cape Leeuwin 00:51:50 (Kadri and ours; bearing 315.5° corrected, 320.4° published). Implied impact 00:29:14–00:37:50 UTC, arc 23; 24° S

Primary site, Cape Leeuwin. Solid red line: onset pick 00:51:50. Dashed line: envelope peak 00:52:10. Red band: arrival span used for timing, 00:51:30–00:52:10.
Envelope evidence p = 0.0958 (smallest within ±15 s of the onset; within-trace null).
Coincides in time with Royer A, figure pick +7 s; Royer A (icequake), table pick -7 s (label only, not removed; Royer's event locations rest on his own unverified pick association).



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

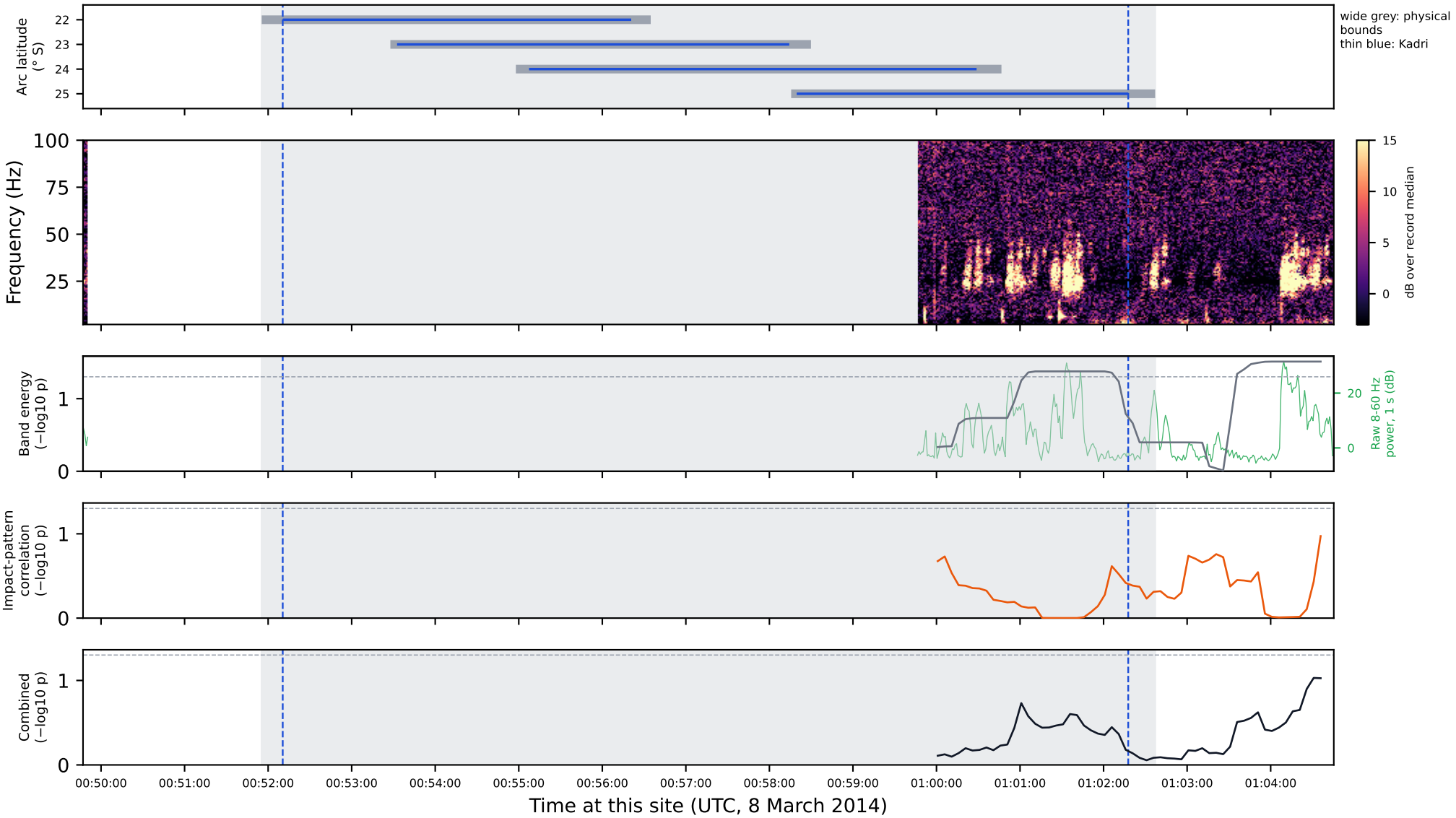
Candidate 5: Cape Leeuwin 00:51:50 (Kadri and ours; bearing 315.5° corrected, 320.4° published). Implied impact 00:29:14–00:37:50 UTC, arc 23; 24° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 5: Cape Leeuwin 00:51:50 (Kadri and ours; bearing 315.5° corrected, 320.4° published). Implied impact 00:29:14–00:37:50 UTC, arc 23; 24° S

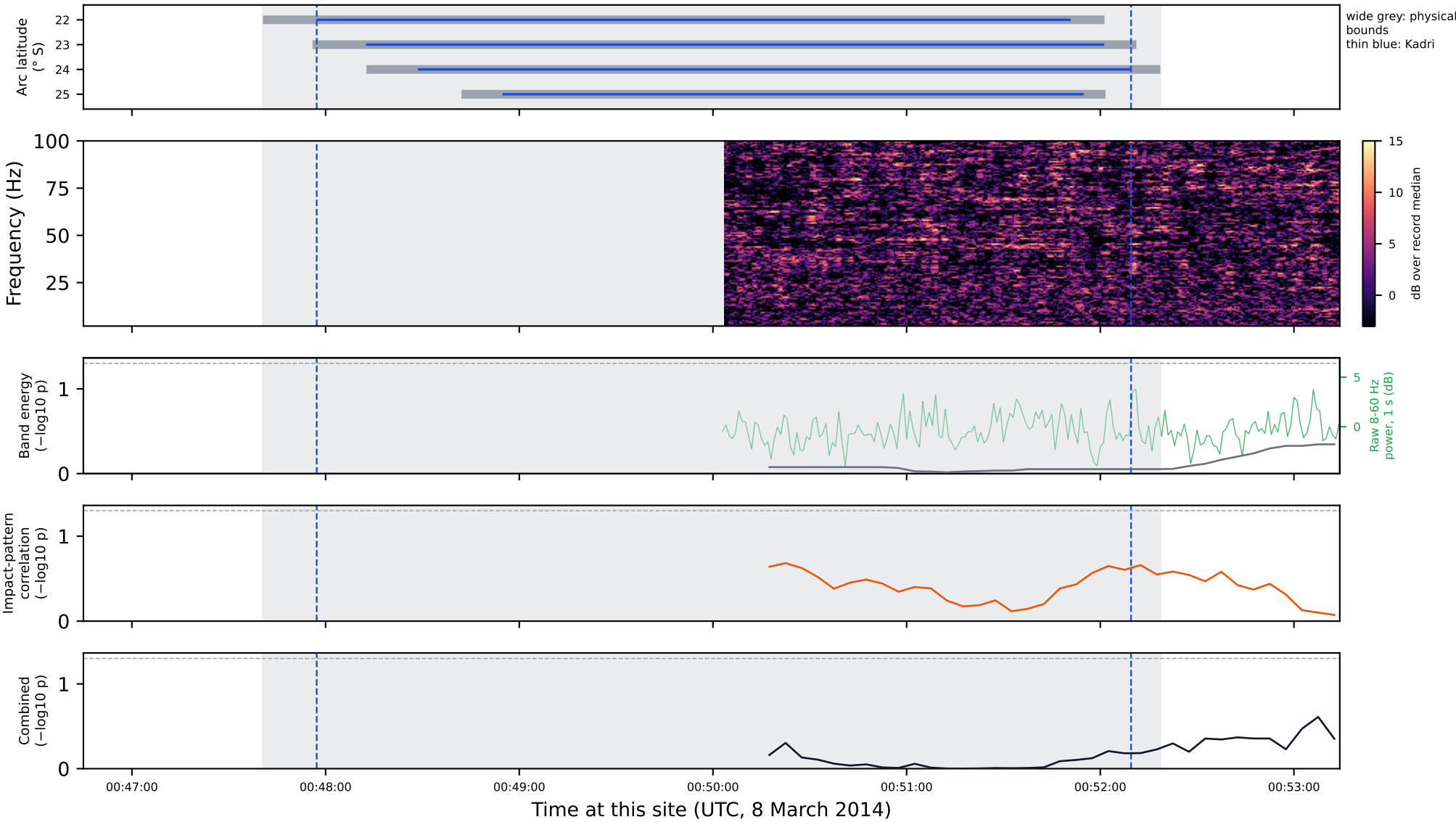
Scott Reef (IMOS 3250). Expected arrival: physical bounds 00:51:55–01:02:37 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:52:10–01:02:17 (blue dashed lines, thin blue bars).
Window shown 00:49:47–01:04:45 (20 % margin). Paths large attenuation (21 of 21 points); 56 looks; window p (combined) = 0.444



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

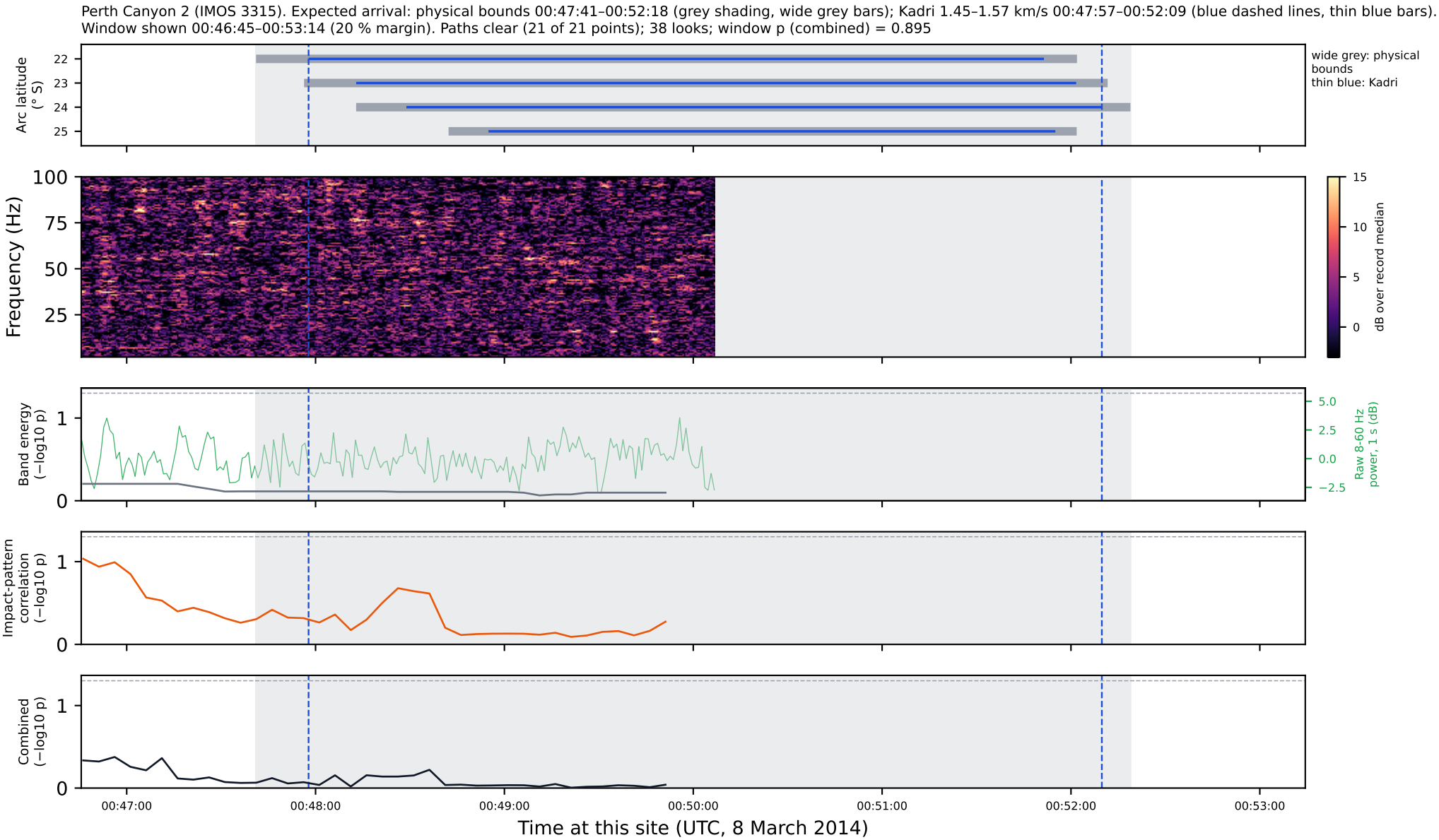
Candidate 5: Cape Leeuwin 00:51:50 (Kadri and ours; bearing 315.5° corrected, 320.4° published). Implied impact 00:29:14–00:37:50 UTC, arc 23; 24° S

Perth Canyon 1 (IMOS 3376). Expected arrival: physical bounds 00:47:40–00:52:18 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:47:57–00:52:09 (blue dashed lines, thin blue bars).
Window shown 00:46:45–00:53:14 (20 % margin). Paths clear (21 of 21 points); 36 looks; window p (combined) = 0.703



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

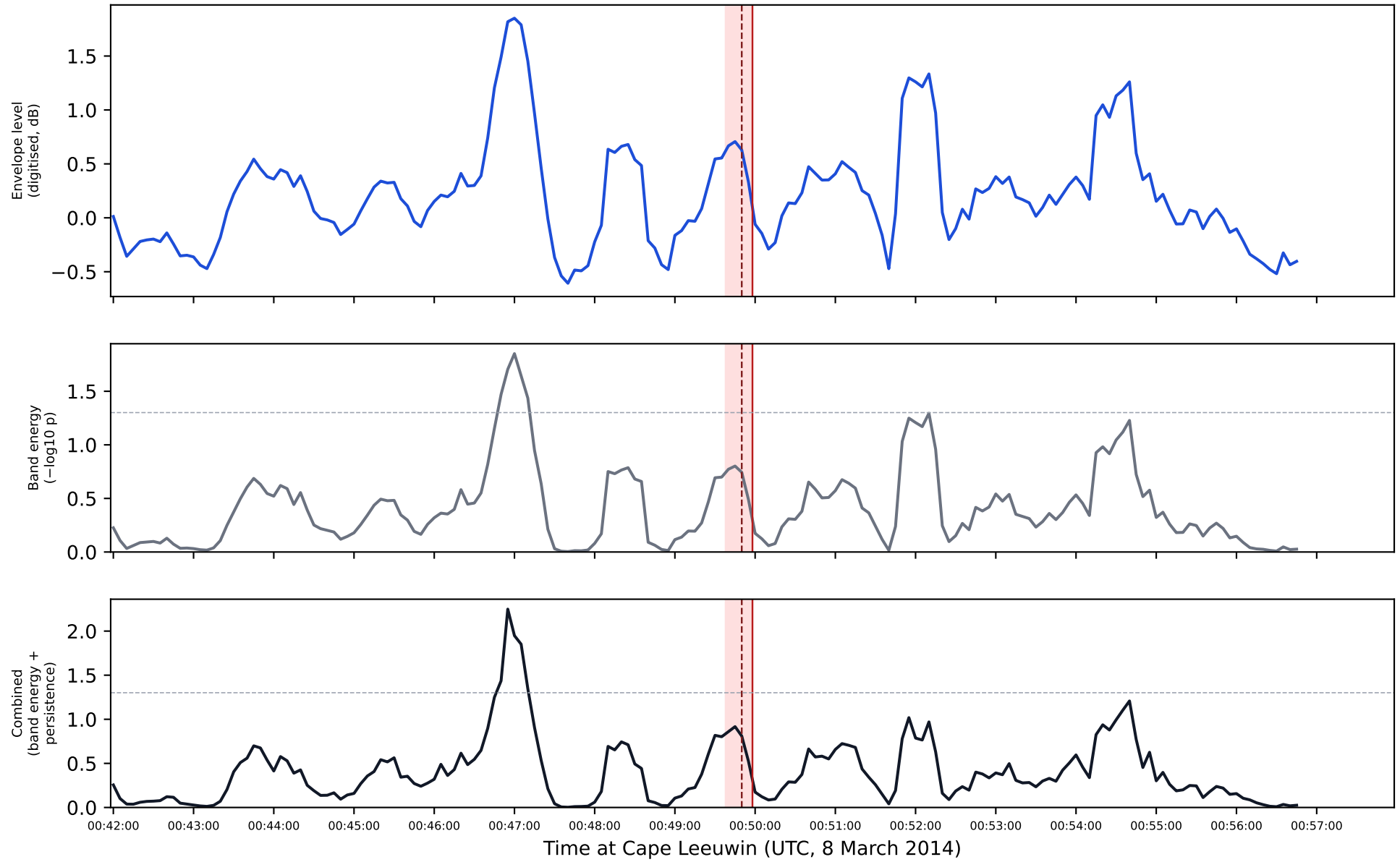
Candidate 5: Cape Leeuwin 00:51:50 (Kadri and ours; bearing 315.5° corrected, 320.4° published). Implied impact 00:29:14–00:37:50 UTC, arc 23; 24° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

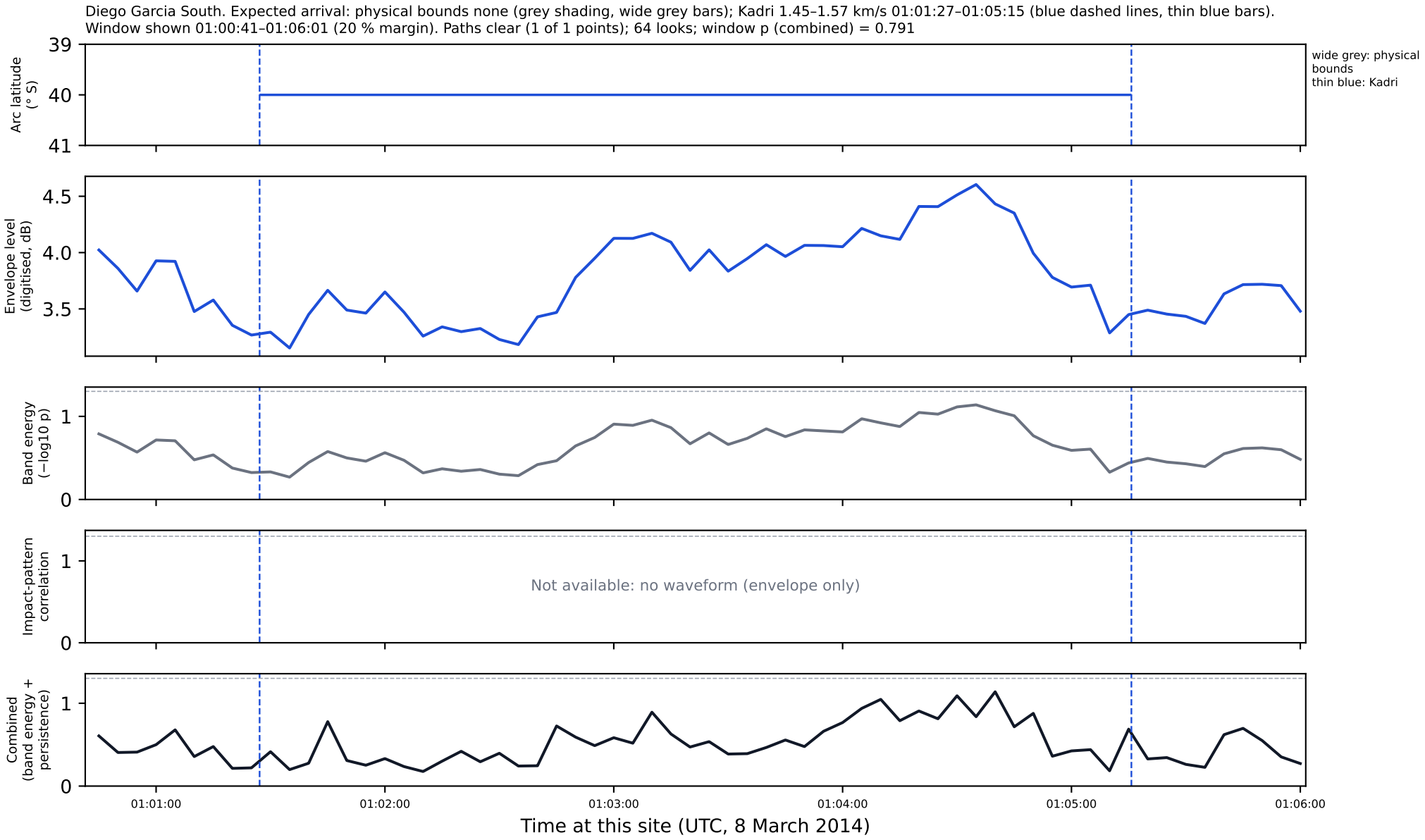
Candidate 6: Cape Leeuwin 00:49:58 (Kadri; bearing 246.8° corrected, 260.4° published). Implied impact 00:20:59–00:21:26 UTC, arc 40° S

Primary site, Cape Leeuwin. Solid red line: onset pick 00:49:58. Dashed line: envelope peak 00:49:50. Red band: arrival span used for timing, 00:49:38–00:49:58.
Envelope evidence p = 0.121 (smallest within ±15 s of the onset; within-trace null).



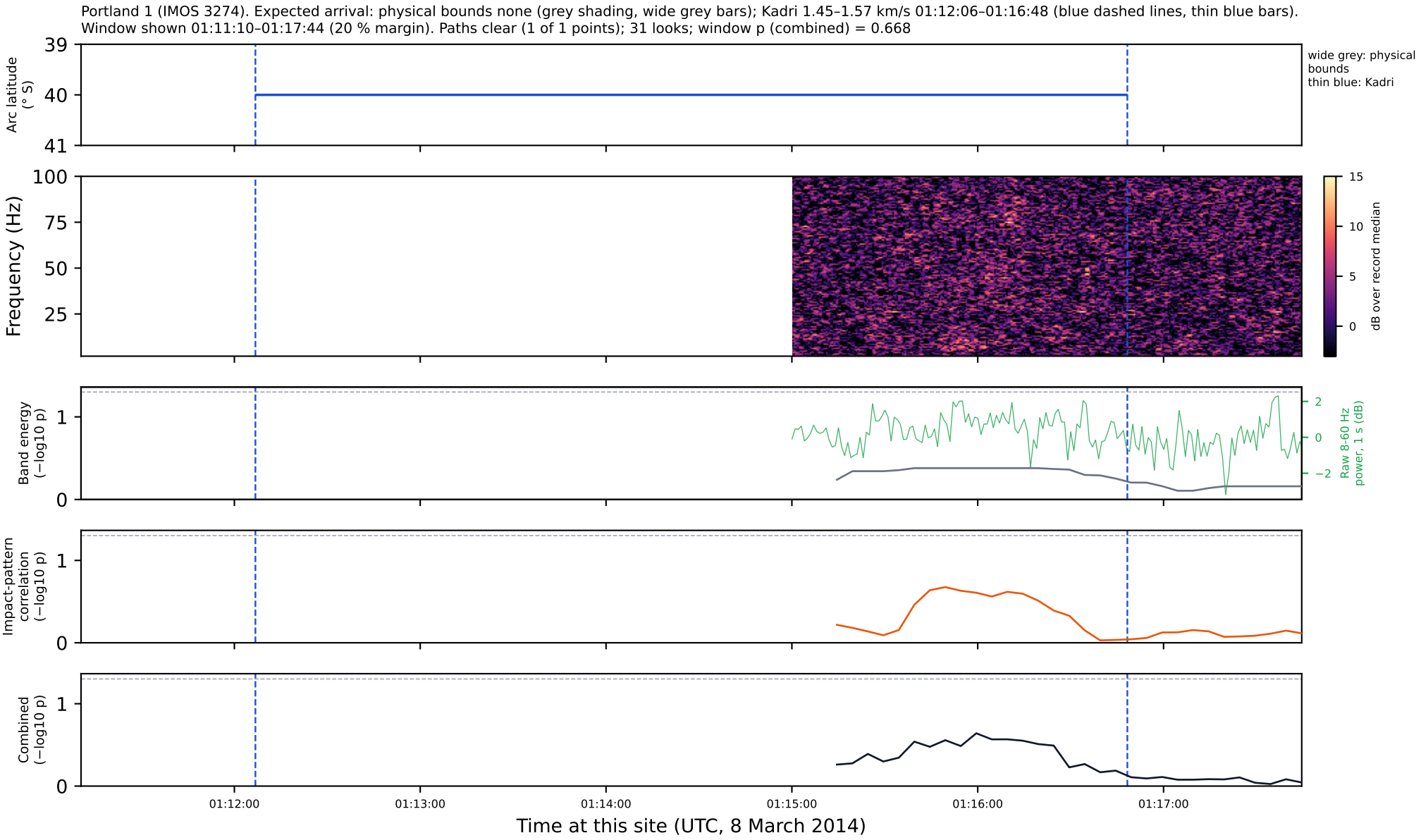
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

Candidate 6: Cape Leeuwin 00:49:58 (Kadri; bearing 246.8° corrected, 260.4° published). Implied impact 00:20:59–00:21:26 UTC, arc 40° S



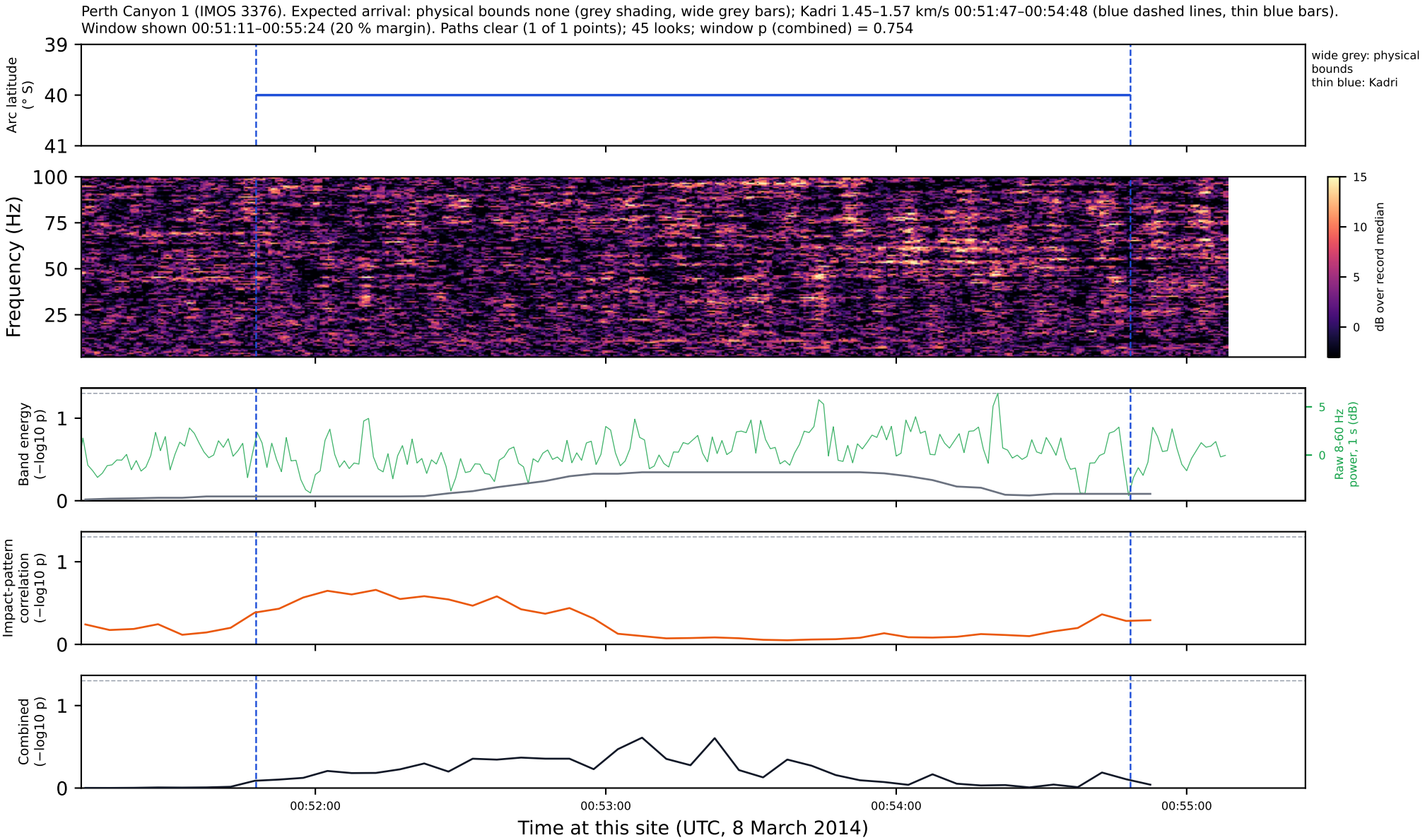
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 6: Cape Leeuwin 00:49:58 (Kadri; bearing 246.8° corrected, 260.4° published). Implied impact 00:20:59–00:21:26 UTC, arc 40° S



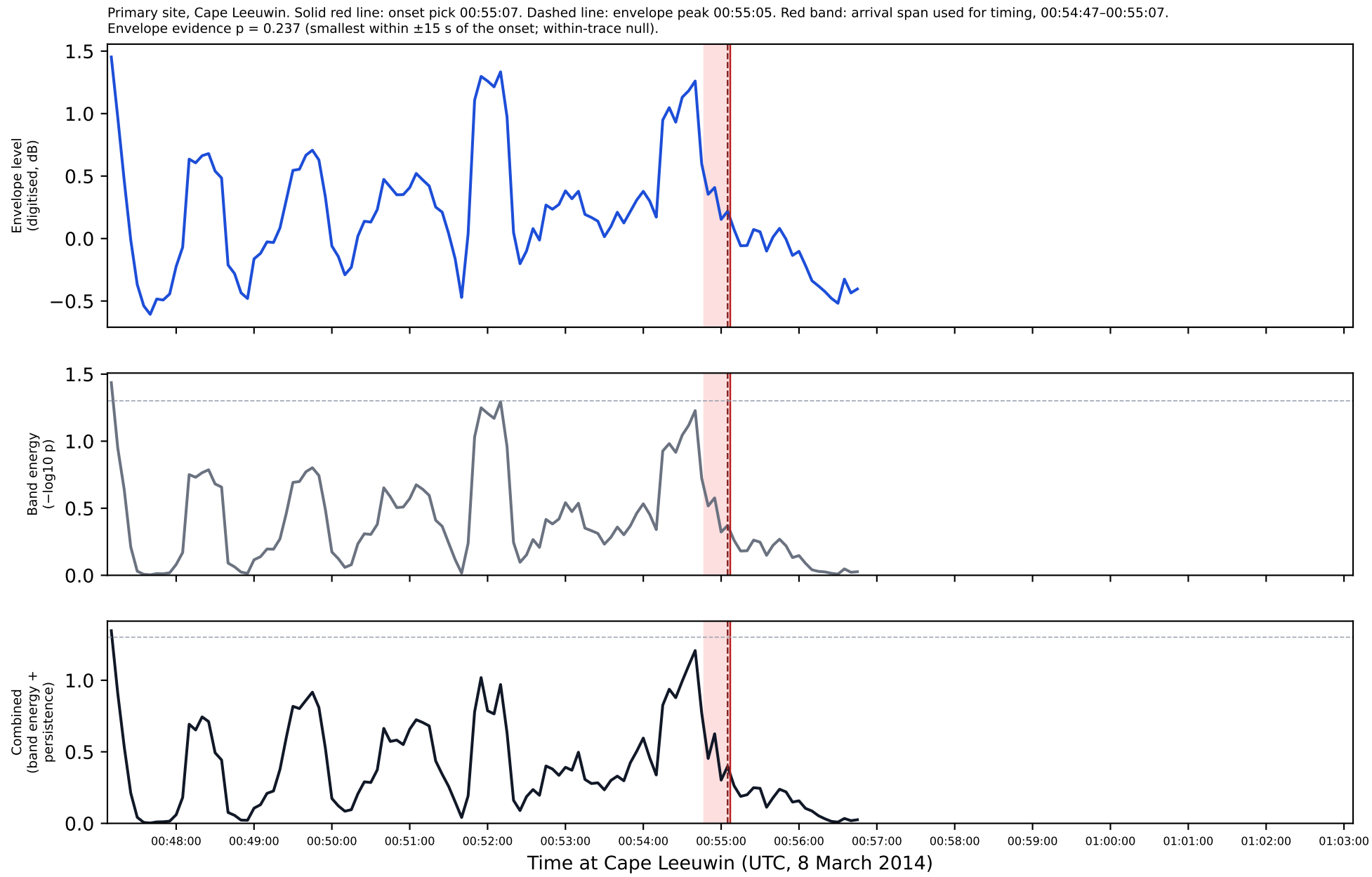
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 6: Cape Leeuwin 00:49:58 (Kadri; bearing 246.8° corrected, 260.4° published). Implied impact 00:20:59–00:21:26 UTC, arc 40° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

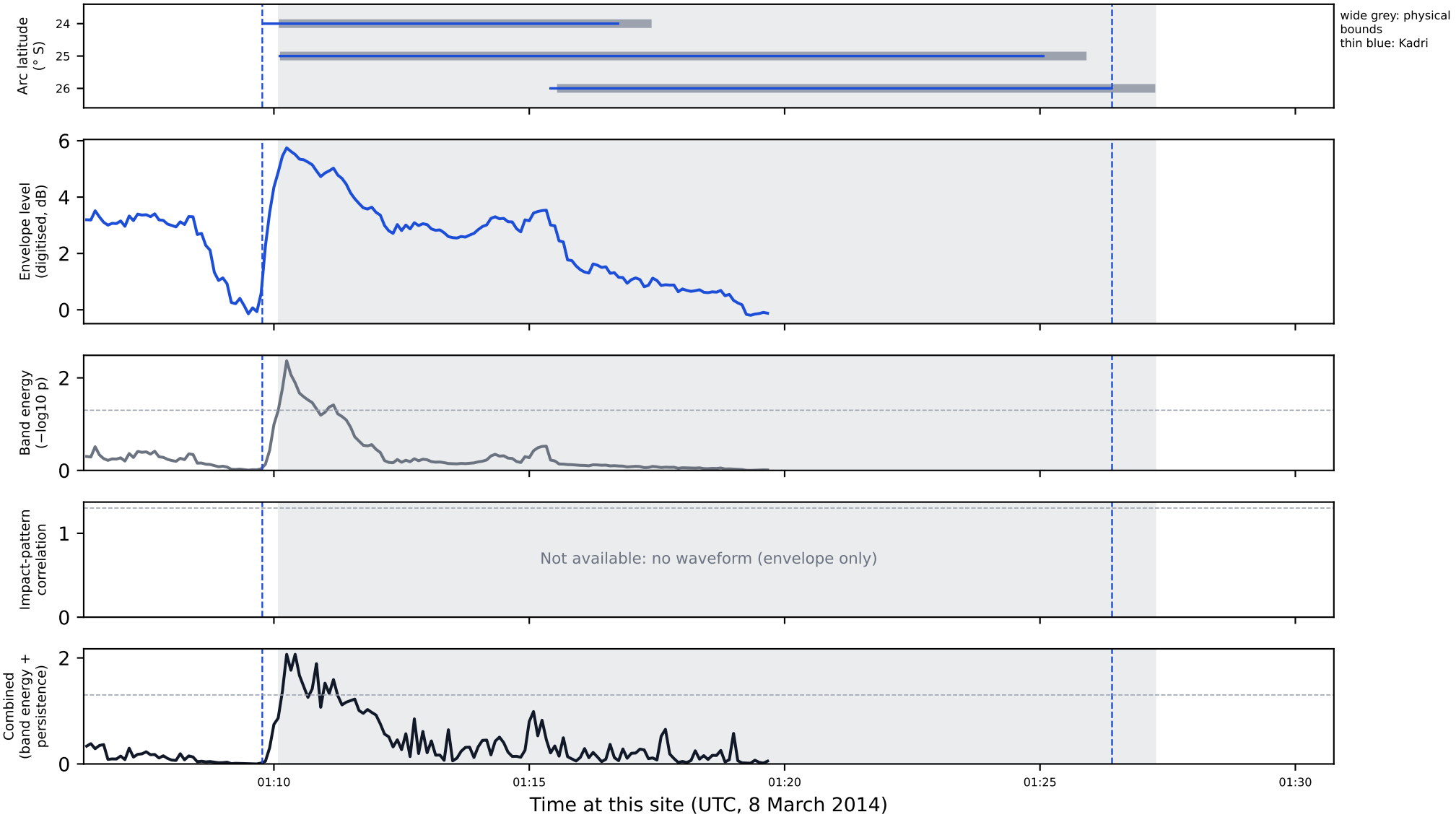
Candidate 7: Cape Leeuwin 00:55:07 (Kadri; bearing 308.5° corrected, 314.1° published). Implied impact 00:32:59–00:41:11 UTC, arc 25; 26° S



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

Candidate 7: Cape Leeuwin 00:55:07 (Kadri; bearing 308.5° corrected, 314.1° published). Implied impact 00:32:59–00:41:11 UTC, arc 25; 26° S

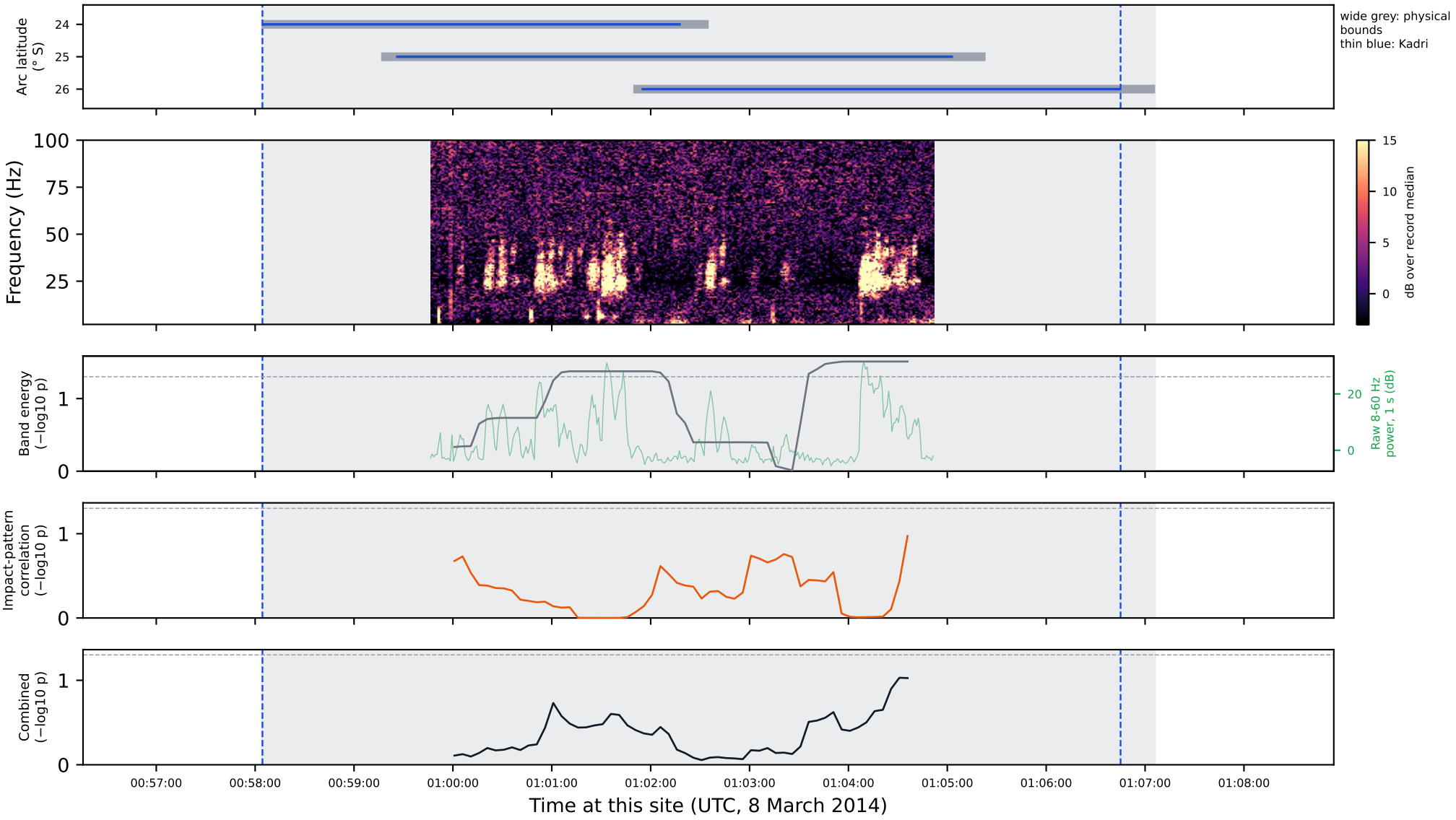
Diego Garcia South. Expected arrival: physical bounds 01:10:05–01:27:15 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 01:09:46–01:26:24 (blue dashed lines, thin blue bars).
Window shown 01:06:16–01:30:45 (20 % margin). Paths clear (21 of 21 points); 161 looks; window p (combined) = 0.749



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 7: Cape Leeuwin 00:55:07 (Kadri; bearing 308.5° corrected, 314.1° published). Implied impact 00:32:59–00:41:11 UTC, arc 25; 26° S

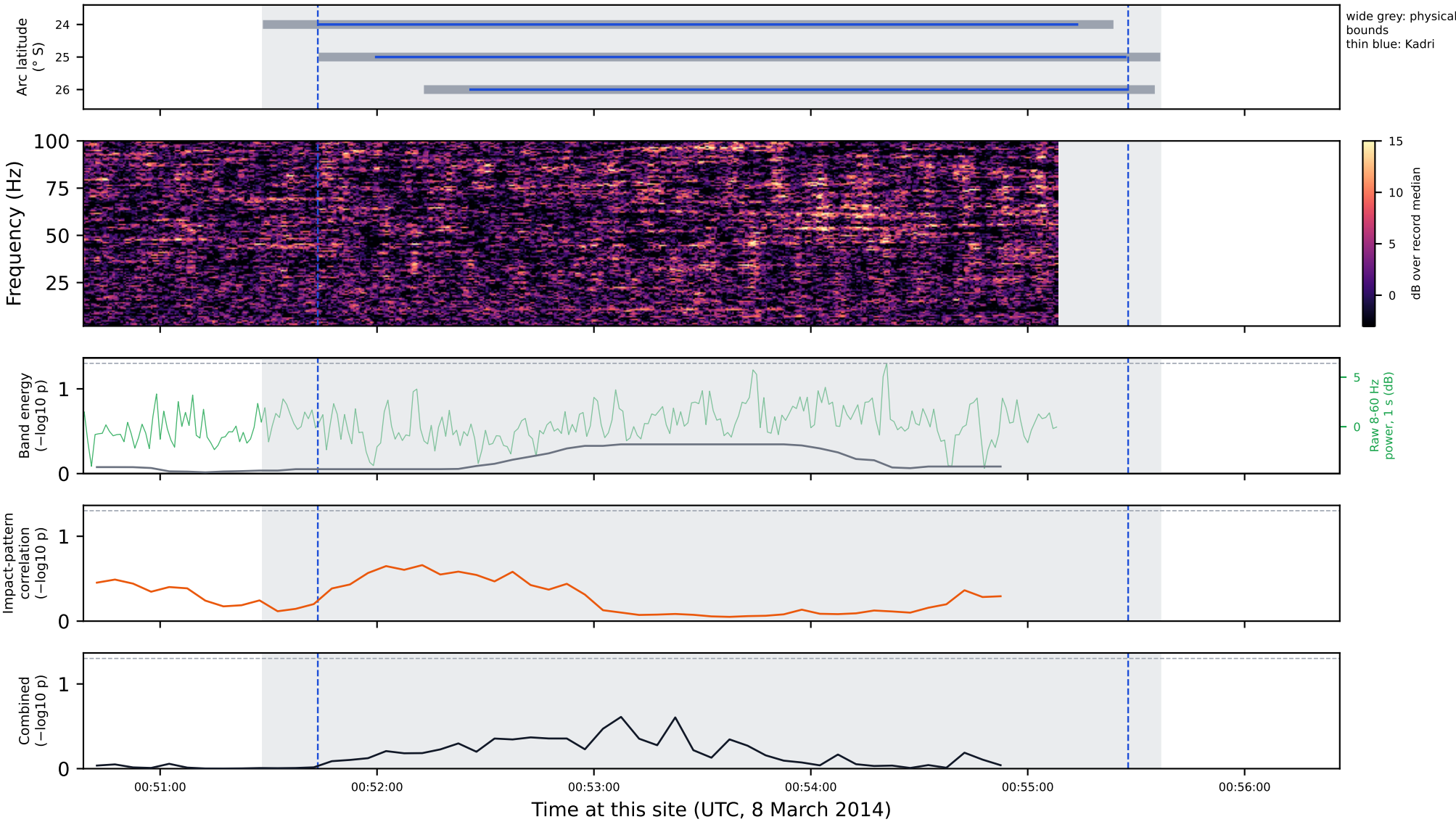
Scott Reef (IMOS 3250). Expected arrival: physical bounds 00:58:04–01:07:06 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:58:04–01:06:45 (blue dashed lines, thin blue bars).
Window shown 00:56:15–01:08:54 (20 % margin). Paths large attenuation (20 of 21 points); 56 looks; window p (combined) = 0.444



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

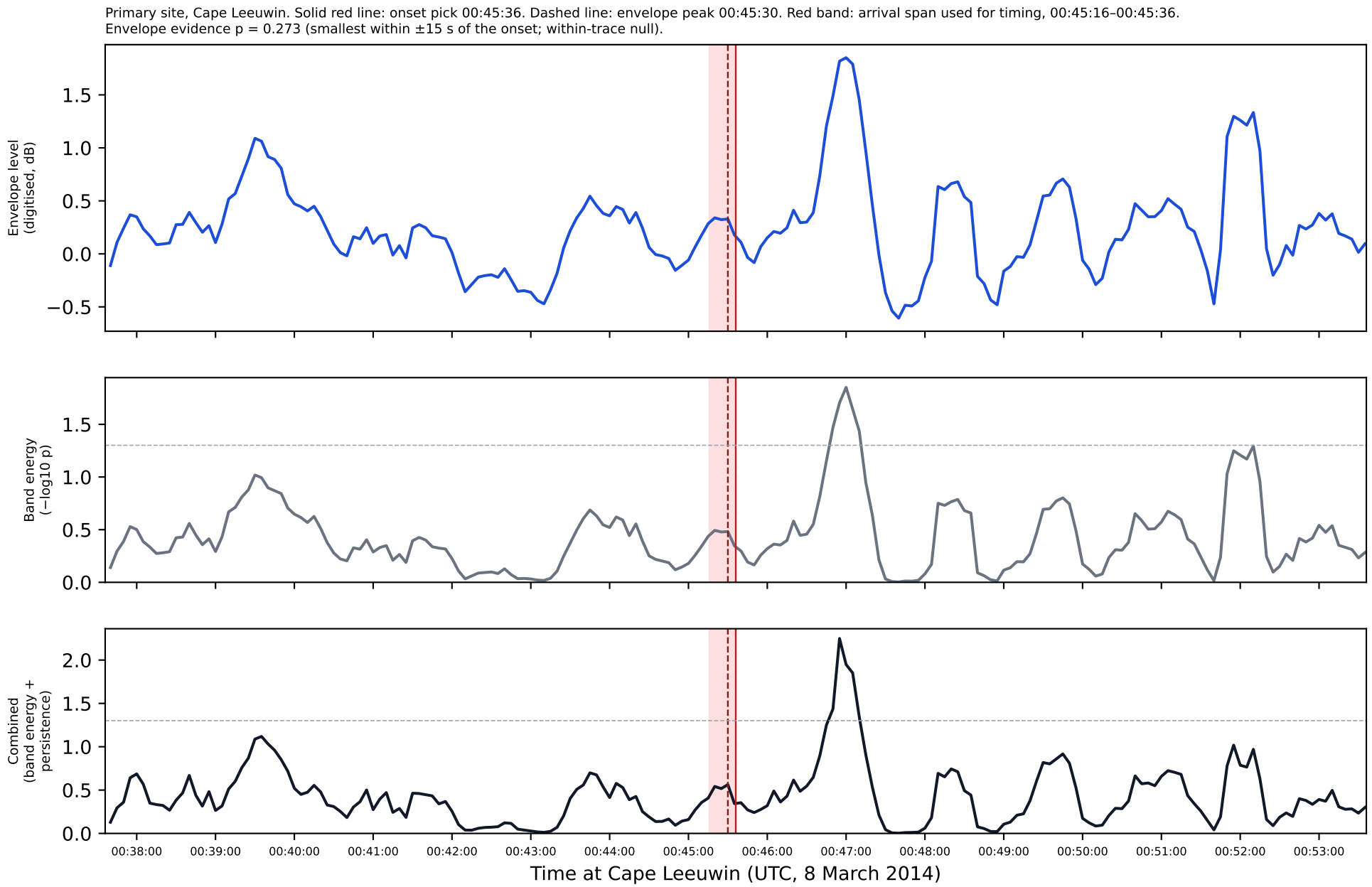
Candidate 7: Cape Leeuwin 00:55:07 (Kadri; bearing 308.5° corrected, 314.1° published). Implied impact 00:32:59–00:41:11 UTC, arc 25; 26° S

Perth Canyon 1 (IMOS 3376). Expected arrival: physical bounds 00:51:28–00:55:36 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:51:43–00:55:27 (blue dashed lines, thin blue bars).
Window shown 00:50:38–00:56:26 (20 % margin). Paths clear (21 of 21 points); 51 looks; window p (combined) = 0.781



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

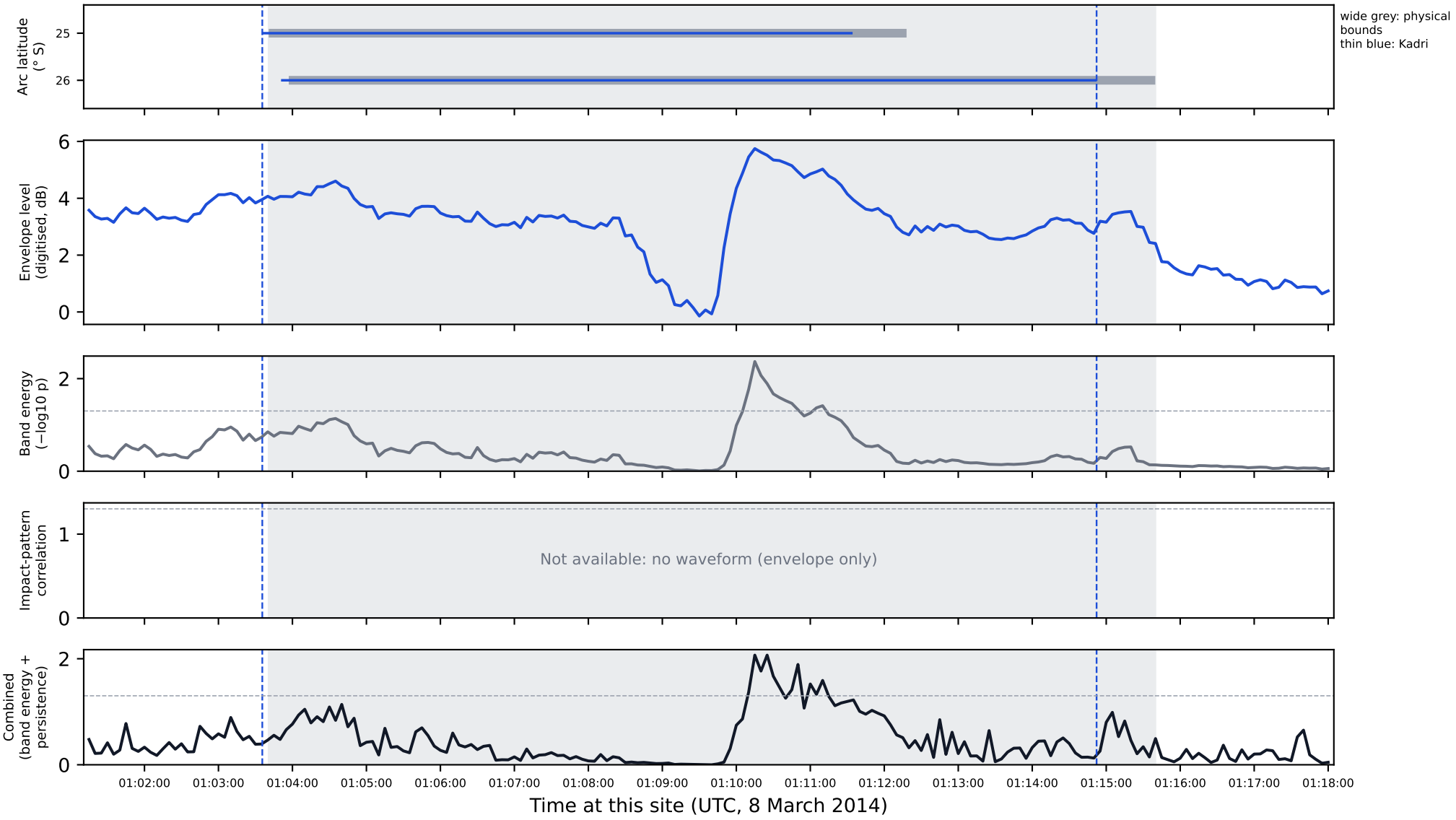
Candidate 8: Cape Leeuwin 00:45:36 (Kadri; bearing 306.1° corrected, 312.0° published). Implied impact 00:25:09–00:30:42 UTC, arc 25; 26° S



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

Candidate 8: Cape Leeuwin 00:45:36 (Kadri; bearing 306.1° corrected, 312.0° published). Implied impact 00:25:09–00:30:42 UTC, arc 25; 26° S

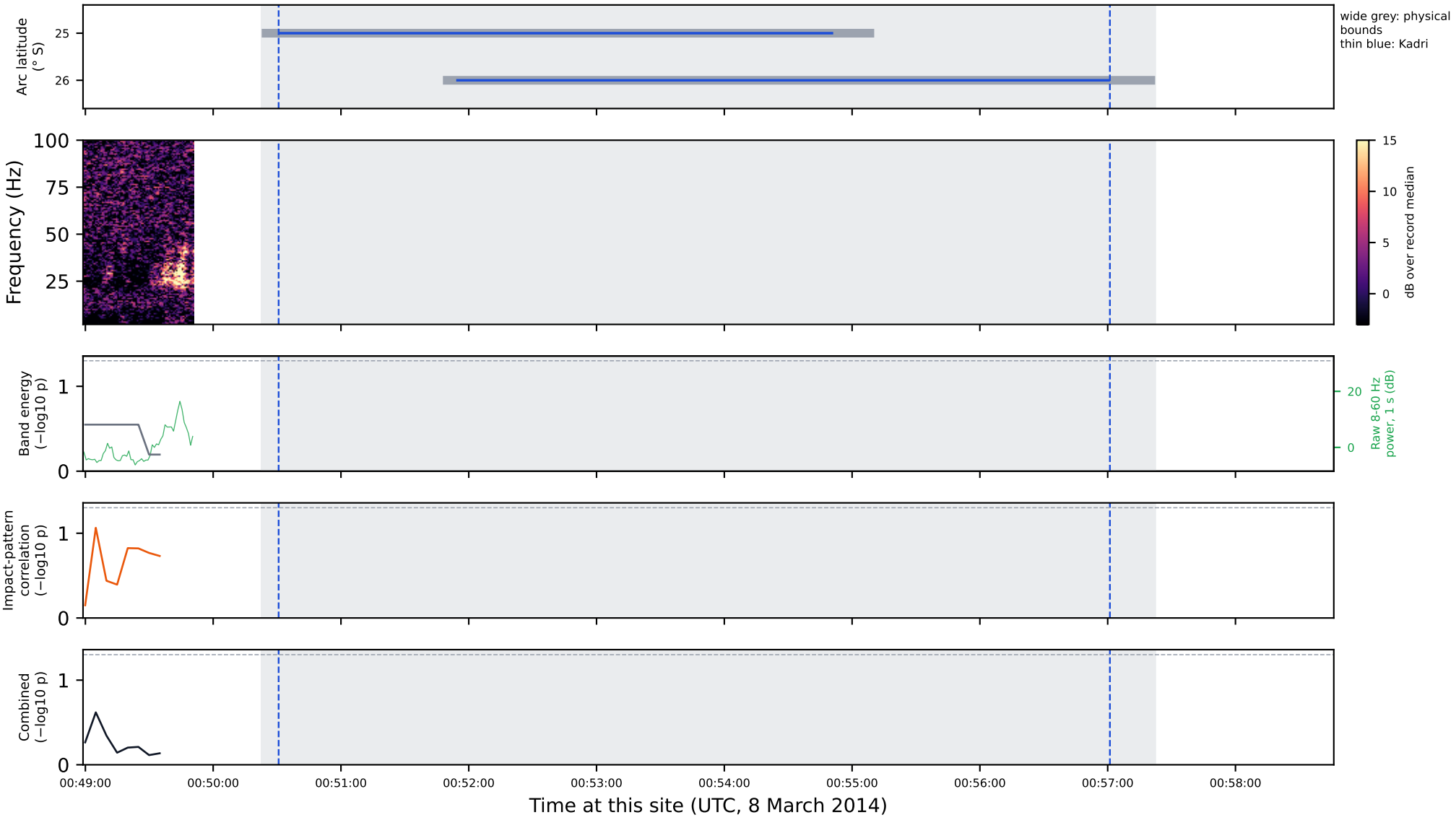
Diego Garcia South. Expected arrival: physical bounds 01:03:40–01:15:39 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 01:03:35–01:14:52 (blue dashed lines, thin blue bars).
Window shown 01:01:10–01:18:04 (20 % margin). Paths clear (13 of 13 points); 202 looks; window p (combined) = 0.749



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

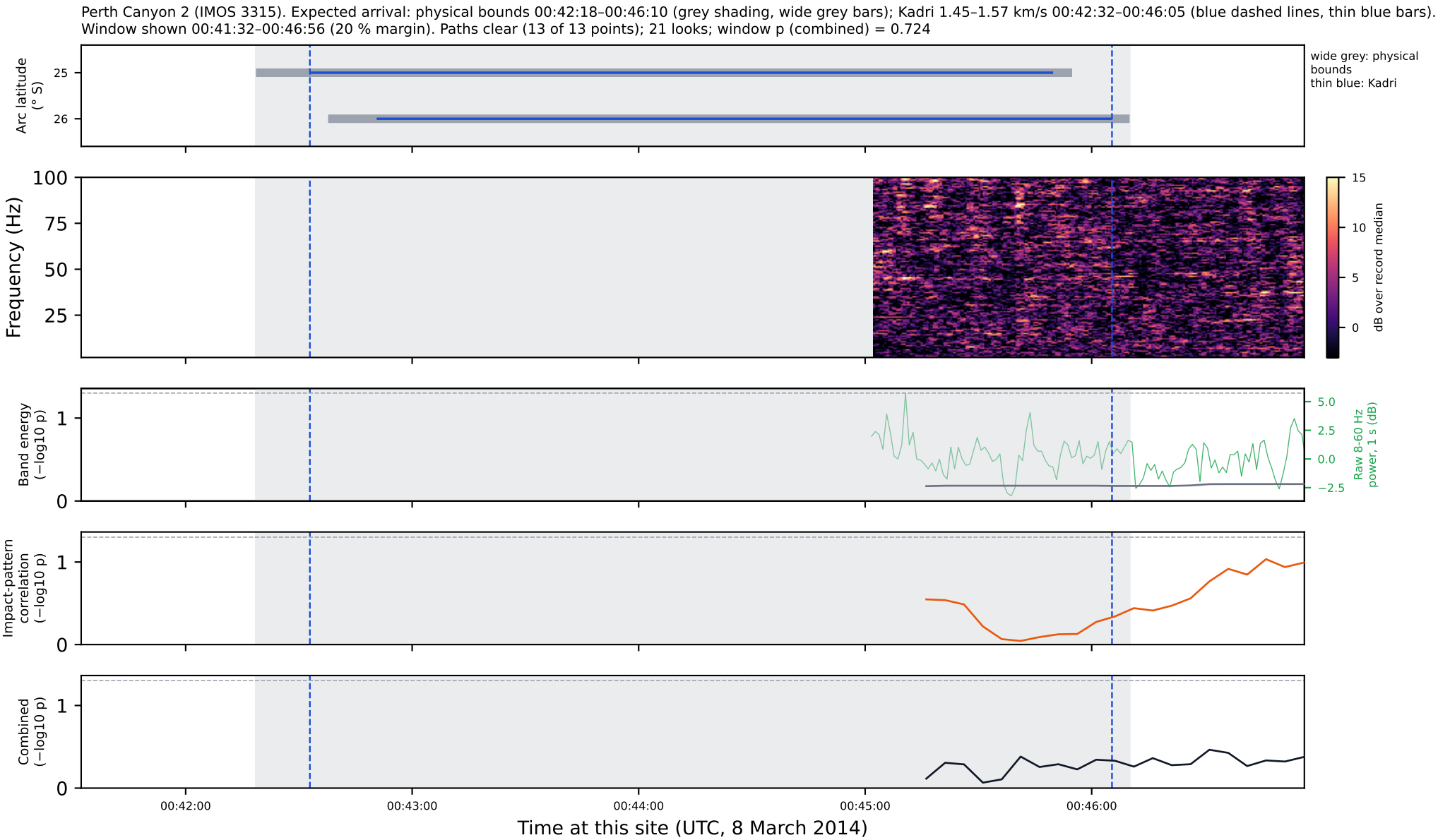
Candidate 8: Cape Leeuwin 00:45:36 (Kadri; bearing 306.1° corrected, 312.0° published). Implied impact 00:25:09–00:30:42 UTC, arc 25; 26° S

Scott Reef (IMOS 3250). Expected arrival: physical bounds 00:50:22–00:57:22 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:50:30–00:57:01 (blue dashed lines, thin blue bars).
Window shown 00:48:58–00:58:46 (20 % margin). Paths large attenuation (13 of 13 points); 8 looks; window p (combined) = 0.379



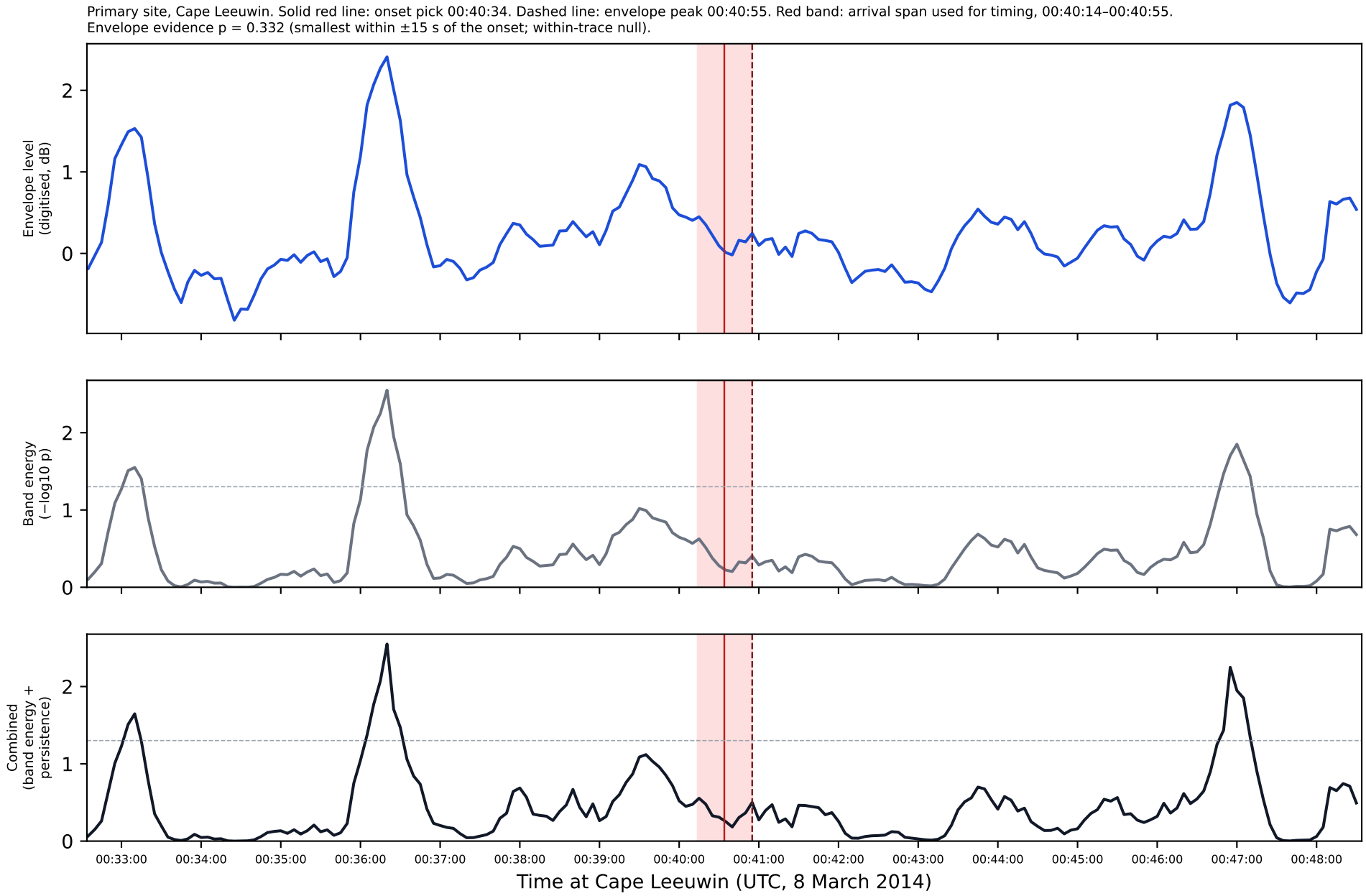
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 8: Cape Leeuwin 00:45:36 (Kadri; bearing 306.1° corrected, 312.0° published). Implied impact 00:25:09–00:30:42 UTC, arc 25; 26° S



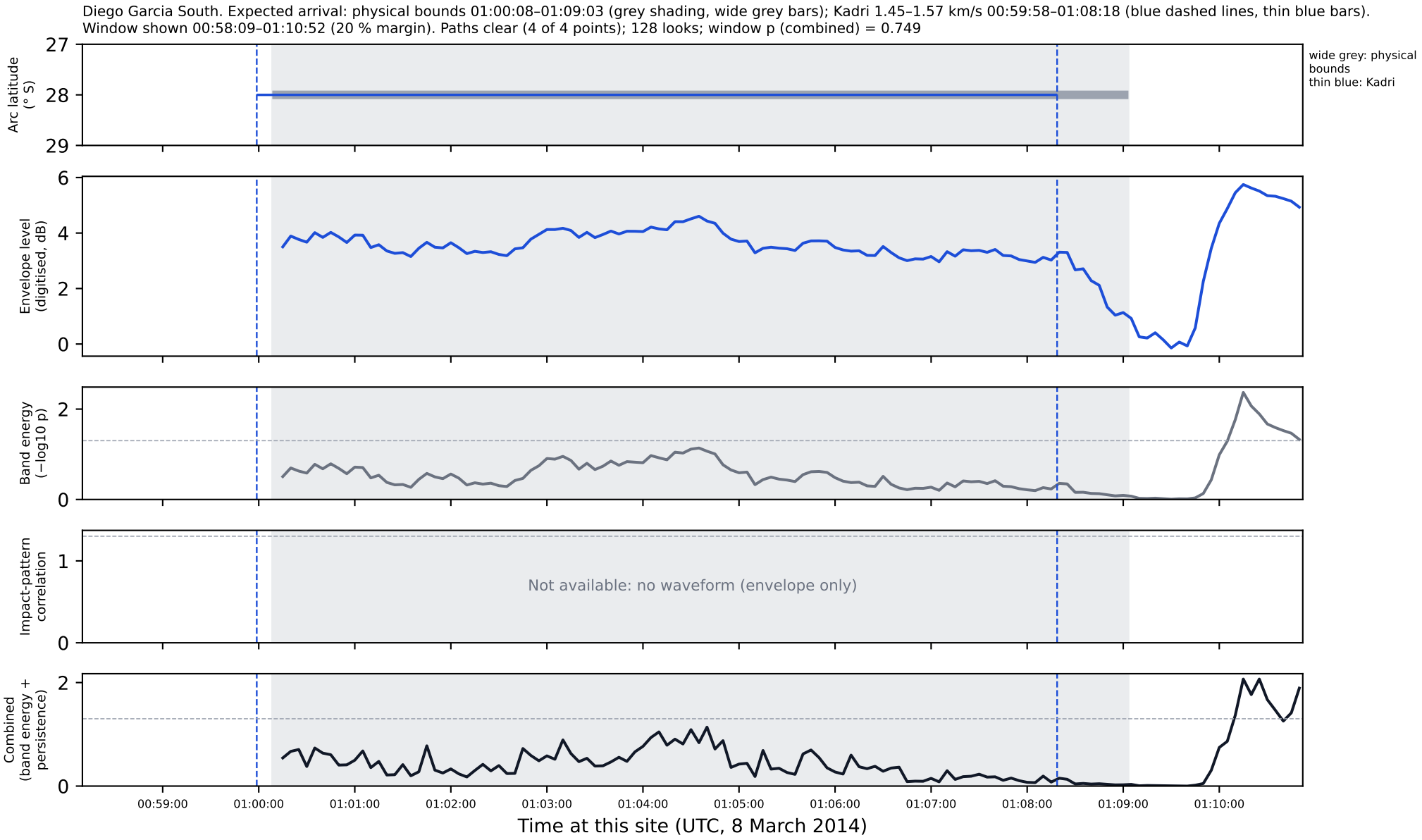
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 9: Cape Leeuwin 00:40:34 (Kadri; bearing 296.0° corrected, 303.3° published). Implied impact 00:21:13–00:25:18 UTC, arc 28° S



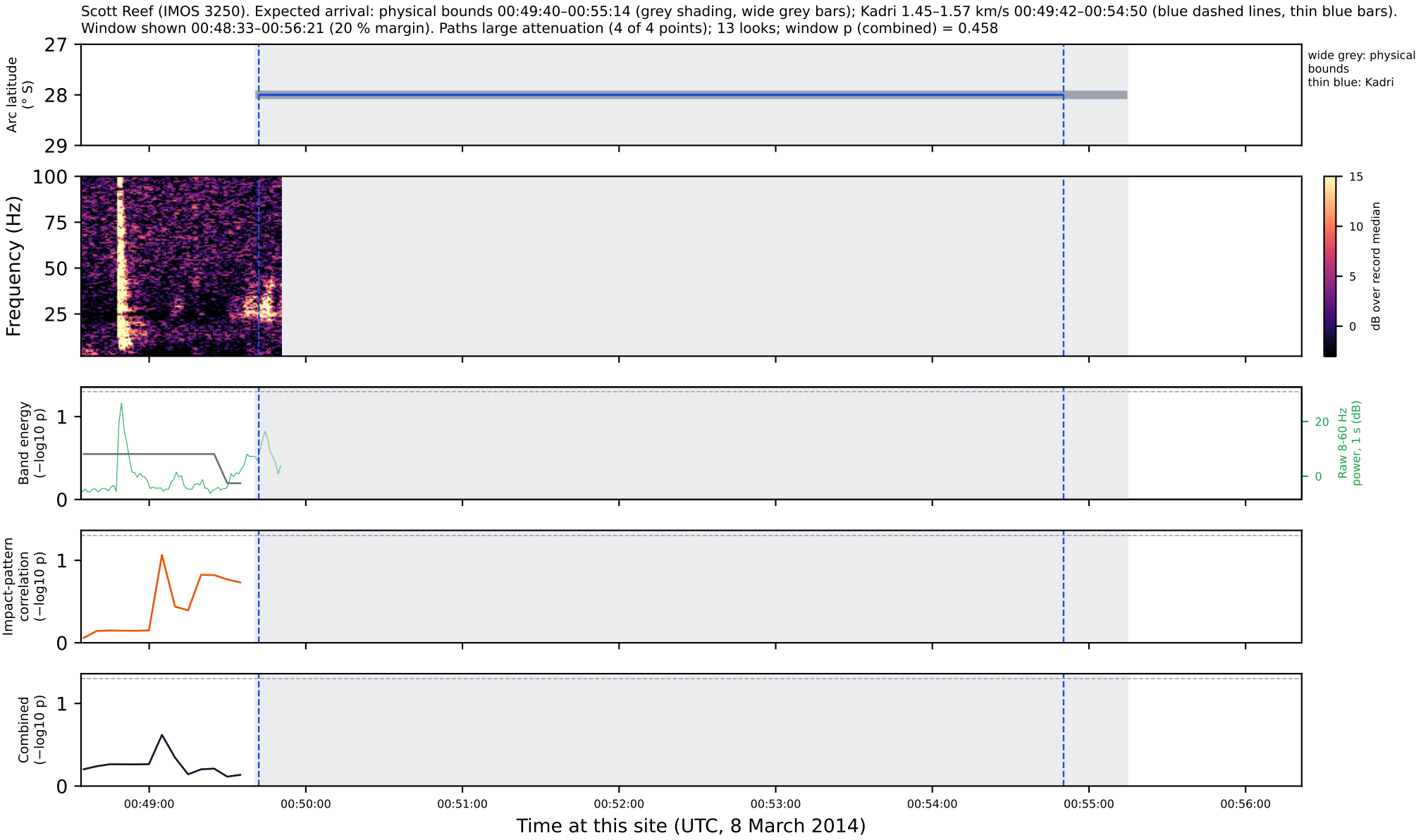
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

Candidate 9: Cape Leeuwin 00:40:34 (Kadri; bearing 296.0° corrected, 303.3° published). Implied impact 00:21:13–00:25:18 UTC, arc 28° S



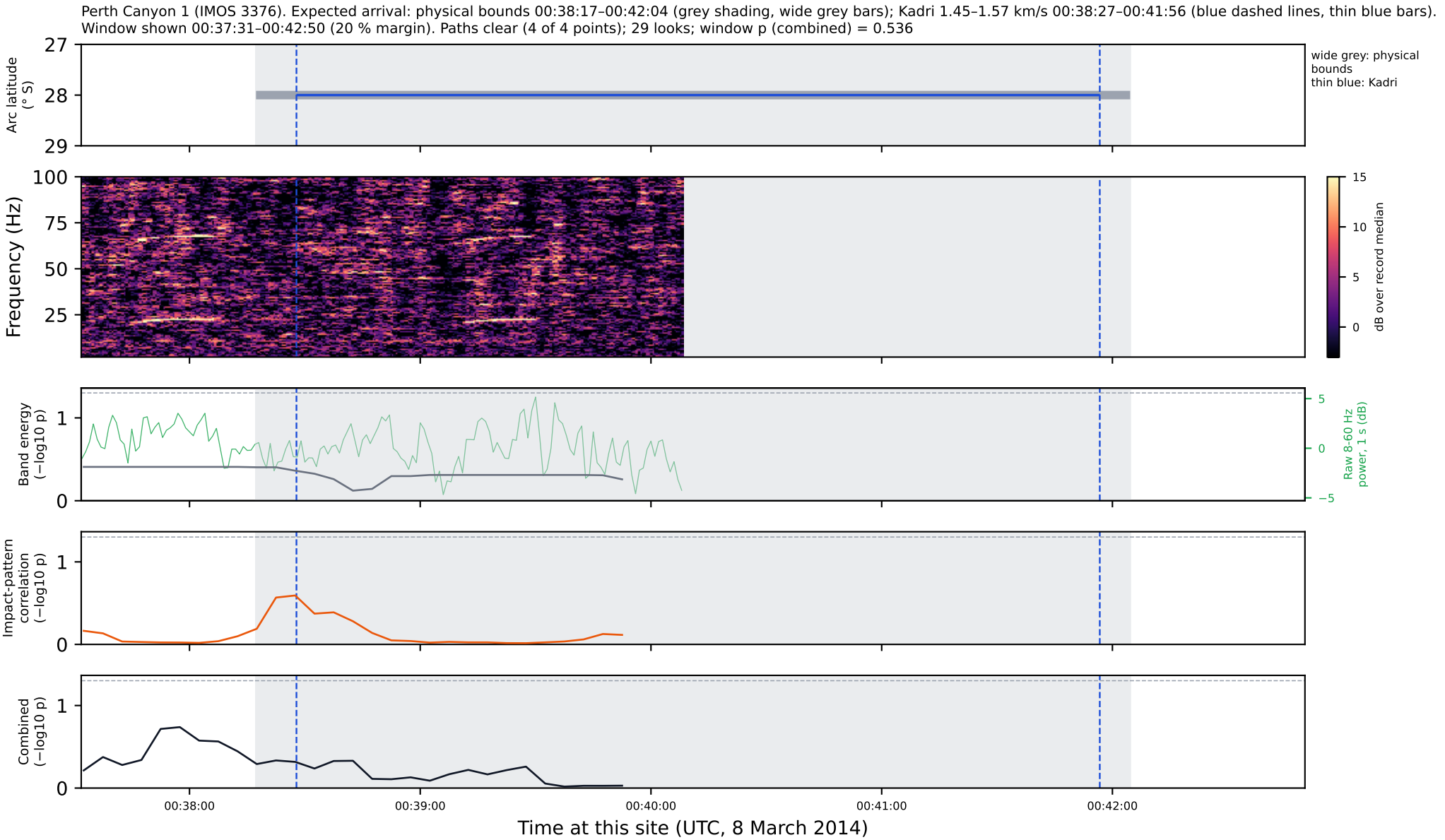
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 9: Cape Leeuwin 00:40:34 (Kadri; bearing 296.0° corrected, 303.3° published). Implied impact 00:21:13–00:25:18 UTC, arc 28° S



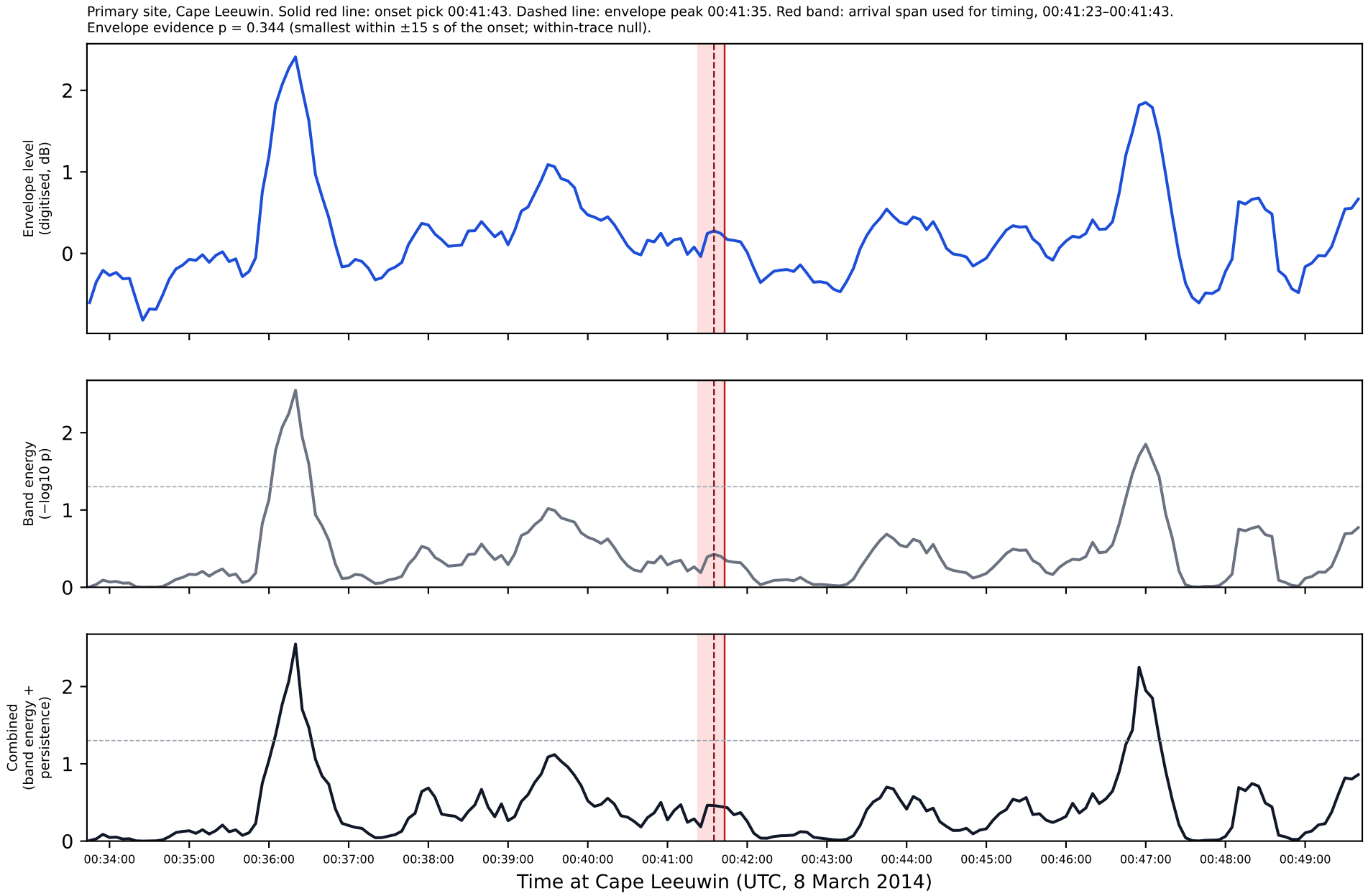
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 9: Cape Leeuwin 00:40:34 (Kadri; bearing 296.0° corrected, 303.3° published). Implied impact 00:21:13–00:25:18 UTC, arc 28° S



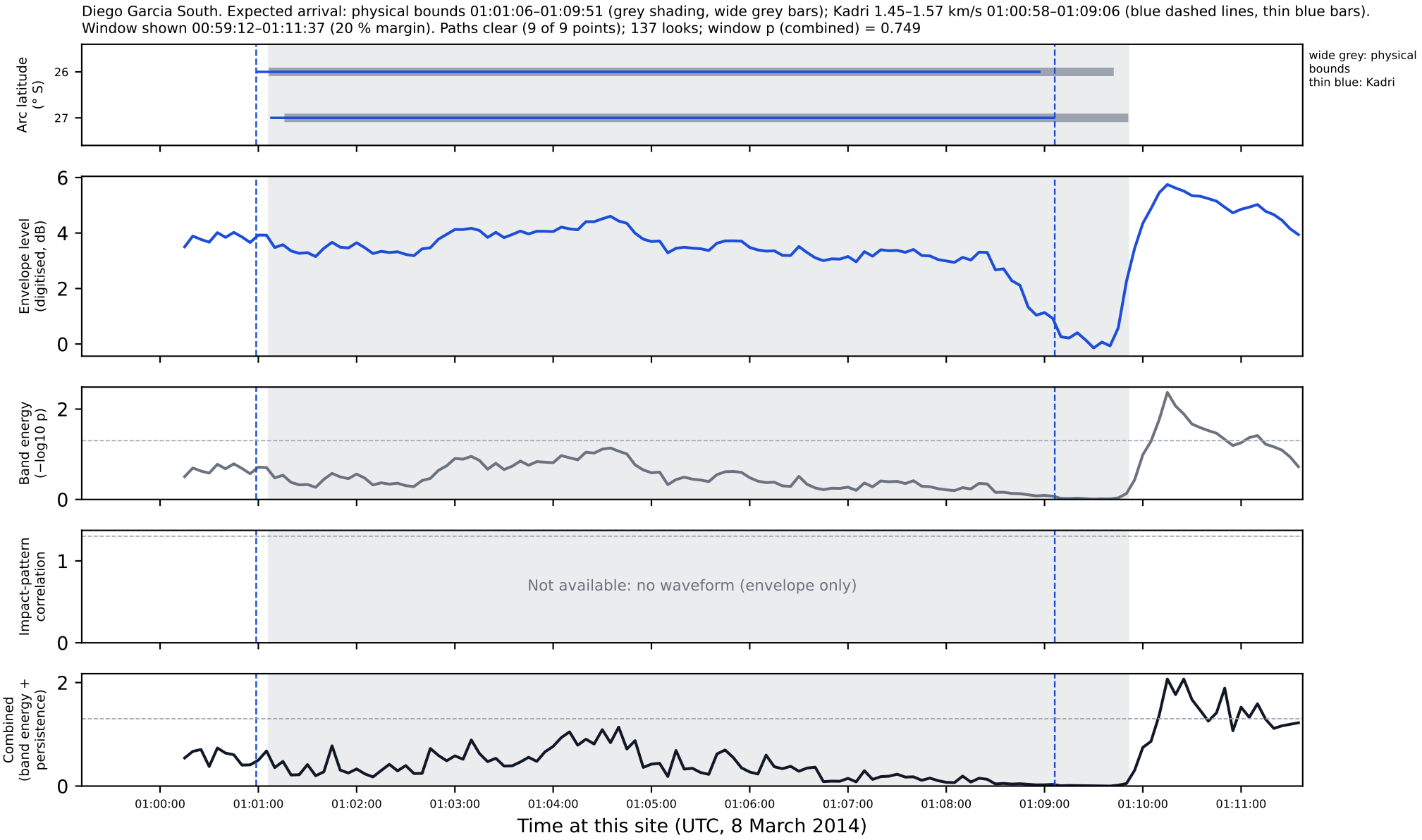
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 10: Cape Leeuwin 00:41:43 (Kadri; bearing 303.8° corrected, 310.0° published). Implied impact 00:22:07–00:26:03 UTC, arc 26; 27° S



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

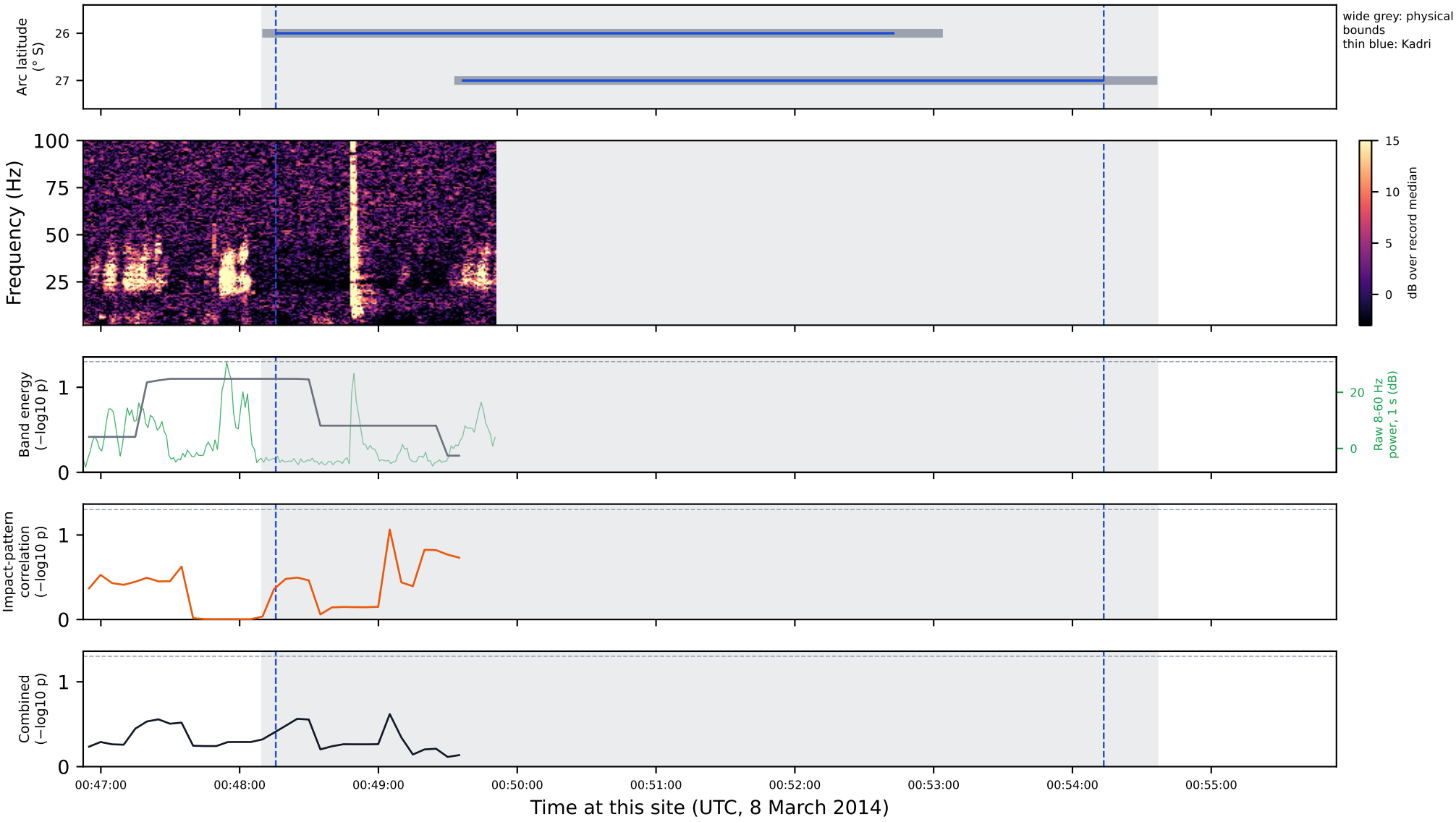
Candidate 10: Cape Leeuwin 00:41:43 (Kadri; bearing 303.8° corrected, 310.0° published). Implied impact 00:22:07–00:26:03 UTC, arc 26; 27° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

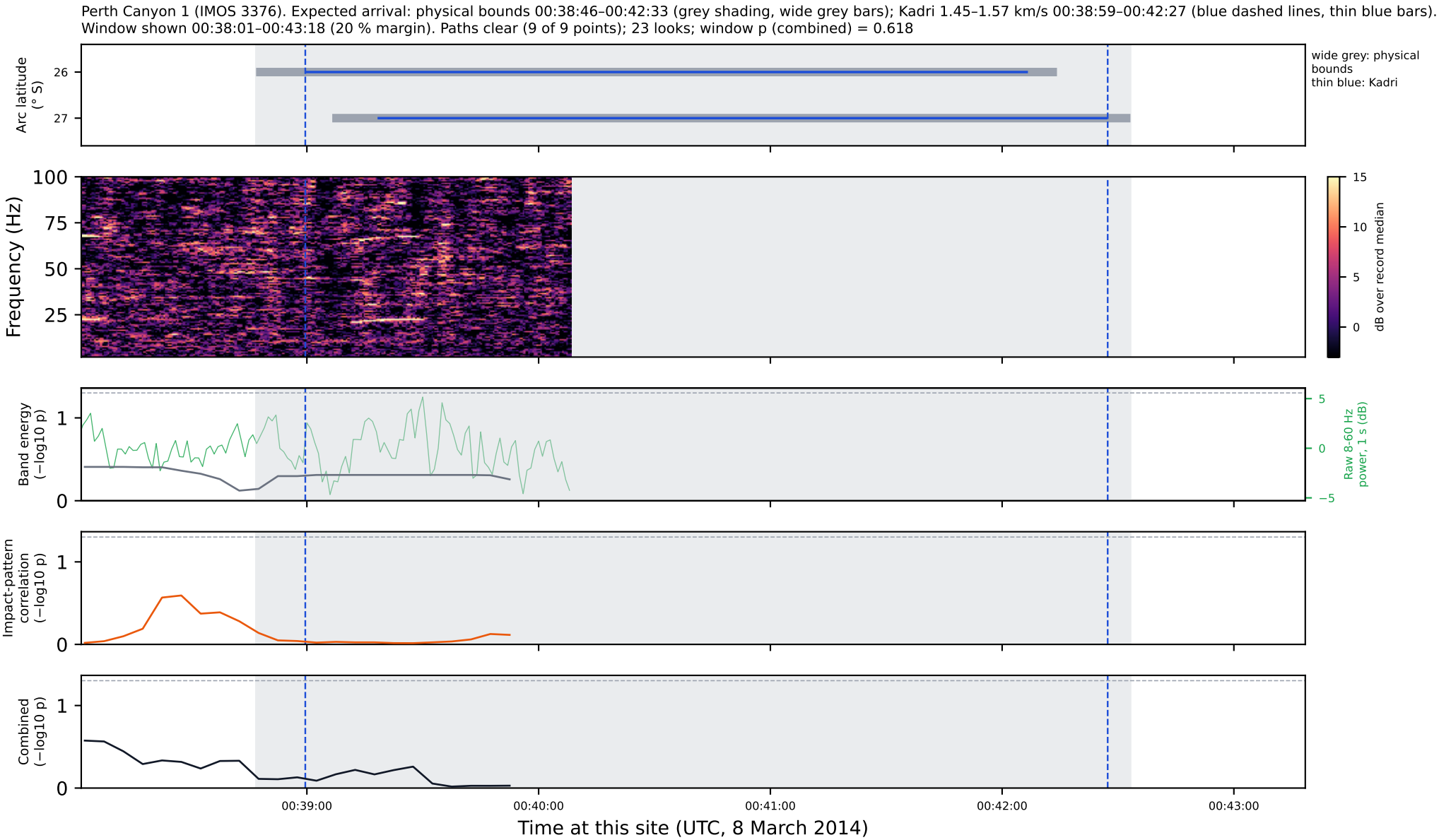
Candidate 10: Cape Leeuwin 00:41:43 (Kadri; bearing 303.8° corrected, 310.0° published). Implied impact 00:22:07–00:26:03 UTC, arc 26; 27° S

Scott Reef (IMOS 3250). Expected arrival: physical bounds 00:48:09–00:54:36 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:48:15–00:54:13 (blue dashed lines, thin blue bars).
Window shown 00:46:52–00:55:54 (20 % margin). Paths large attenuation (9 of 9 points); 33 looks; window p (combined) = 0.674



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey shading and wide grey bars: physical bounds for sound carried in the water on each path; blue dashed lines and thin blue bars: Kadri's window. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate and reachable by the aircraft (no more than 500 kt over ground after the 7th arc at 00:19:29, plus 12.5 NM for the grid spacing; corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

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